The center and the ends of the elephant random walk with rare reversals

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Abstract

In the elephant random walk each step copies a uniformly chosen earlier step and reverses it with probability ε. We let $\varepsilon \to 0$ as the number N of steps grows and ask when the walk is as likely to end at the origin as at N. For even N we prove that this crossing is unique and that $N\varepsilon$ there equals $2\log N - \log\log N - \log 16 + o(1)$, about twice the value for the persistent random walk. Explicit two-sided bounds show that the walk ends at the origin with probability $4\varepsilon/(N+2)$ up to a factor $1 + O(\varepsilon \log N)$. To first order in ε, a walk that ends at the origin reverses once, within its first few steps. At the crossing, the number of steps that disagree with the first step is within an explicit distance of a compound Poisson law of Luria–Delbrück type, and after centering and scaling it obeys a local limit theorem with the Landau law, with explicit constants. We locate a hump of the law of the final position near each of $\pm N$. For every N and every $\varepsilon < \frac{1}{2}$ the second difference of this law changes sign at most four times. For every large even N, at the crossing, the origin is less likely than every other position the walk can reach except $\pm N$, and a computer-assisted proof gives the same for every even N from 8 to 2000. The even N above 2000 and below the threshold for large N, which we do not compute, remain open.

Keywords: elephant random walk; random recursive tree; broadcasting on trees; compound Poisson approximation; Landau distribution; computer-assisted proof.
Mathematics Subject Classification (2020): Primary 60G50; Secondary 60E07, 60C05, 60F05, 60J10, 05C05, 05C80, 65G30.

1 Introduction

The elephant random walk of Schütz and Trimper [1] has steps $X_1, \dots, X_N \in \{\pm 1\}$ and remembers its whole past. Each step after the first picks one of the earlier steps uniformly at random and repeats it, except that with probability ε it reverses it. A reversal reverses the sign of the sampled step, which need not be the step just before. In [1] the first step is $+1$ with a probability that is a parameter of the model, and we take it to be $\frac{1}{2}$. For $\varepsilon \le \frac{1}{2}$ there is an equivalent description, in which a step copies the sampled step exactly with probability $1 - 2\varepsilon$ and is otherwise a fresh fair sign. The steps are also the colors of broadcasting on a random recursive tree [2, 3], in which each new vertex attaches to a uniform earlier vertex, copies its color and changes it with probability ε (Lemma 3.1).

Most limit theorems for the walk fix ε as $N \to \infty$ [4, 5, 6, 7, 8]. For a walk with N steps we write $x = N\varepsilon$. The expected number of reversals is $(N-1)\varepsilon = x - \varepsilon$, so x is this number up to an error below one. We study the regime in which x is of order $\log N$, so that $\varepsilon \to 0$ and the walk reverses only about x times, and we describe the law of its endpoint site by site.

The parameters of other papers translate as follows. The memory parameter of [1] is $1-\varepsilon$, the parameter a of [7] is $1-2\varepsilon$, and the flip probability of broadcasting is ε. For $\varepsilon \le \frac{1}{2}$ the retention probability of the percolation in Section 1.3 is also $1-2\varepsilon$.

We ask when the walk is as likely to end in the middle as at each end. Let N be even, and let A_N be the number of steps equal to $+1$. The walk ends at the origin, which we call the middle, when $A_N = N/2$. It ends at N or at $-N$, which we call the ends, when $A_N = N$ or $A_N = 0$. If ε is close to 0 the walk rarely reverses, so the ends win. If $\varepsilon = \frac{1}{2}$ the steps are independent and fair, so the middle wins. The crossing is the value of ε in between at which the middle and each end are equally likely. In the language of broadcasting, the walk ends in the middle when the two colors tie.

In the persistent random walk each step repeats the previous one with probability p and switches with probability $q = 1-p$. So with $q = \varepsilon$ it is the walk that copies its last step instead of a uniformly chosen one. Its steps are the spins of the Ising chain in zero field with free ends, and its position is the magnetization. For this Markov walk our question is part of Problem 131 of Kagey's Open Problem Collection [9], which is posed for a Galton board. As on that board, we call the values of A_N bins. Let $T = Nq$, which exceeds the expected number $(N-1)q$ of switches by q. The middle catches up with the ends when T is about $\log N$, and [10] locates this crossing within explicit bounds for every $N \ge 2$.

In this paper we find the crossing of the elephant walk. The middle needs about twice as many reversals as the persistent walk needs switches, and the crossing value of x is about $2\log N$, not $\log N$. Kürsten's coupling [11] splits the steps into the clusters of bond percolation on a random recursive tree, and our regime $x \asymp \log N$ lies in the strongly supercritical regime of this percolation [12]. The results there describe cluster sizes, not the probability of the middle (Section 1.3).

The number of steps that disagree with the first step is a known chain. With equal switching probabilities in both directions, it is the number of resistant cells in the reversible fluctuation assay of Bokes, Hlubinová and Singh [19]. They find its law to first order in the switching probability by conditioning on the first switch and then running Pólya's urn [19, Eqs. (6)–(10)]. At the center this gives our first-order coefficient $4/(N+2)$. For a first cell of random type they also approximate the law by a mixture of two Landau densities, one at each end [19, Eqs. (58)–(60)]. Their analysis uses matched asymptotics and gives no rigorous error estimates. What we add are rigorous estimates of the center with uniform relative errors while x grows with N, the expansion of the crossing, and theorems on the discrete shape of the law.

1.1 Setting and notation

The walk takes N steps $X_1, \dots, X_N \in \{+1, -1\}$, where $+1$ means right. Let $A_N = \#\{i \le N : X_i = +1\}$ be the number of right steps, and write $P_N(k) = \mathbb{P}(A_N = k)$ for the probability of bin k. Unless we say otherwise the first step is fair, $\mathbb{P}(X_1 = +1) = \frac{1}{2}$, and the rules treat the two directions alike, so $P_N(k) = P_N(N-k)$. We write $\mathbb{P}_+$ and $\mathbb{E}_+$ for probability and expectation given $X_1 = +1$. For even N we put $h = N/2$ and call $C_N = P_N(h)$ the center and $E_N = P_N(N) = P_N(0)$ the end. Whenever C_N appears, N is even. The crossing is the value of the parameter at which $C_N = E_N$. All logarithms are natural, $L = \log N$, $H_k = \sum_{j \le k} 1/j$ is the k-th harmonic number,

and γ is Euler's constant. Limits, $O(\cdot)$ and $o(\cdot)$ refer to $N \to \infty$, over even N when C_N appears. The constants implied by $O(\cdot)$ are absolute unless we name what they depend on.

The elephant walk (Definition 2.1) has the reversal probability $\varepsilon \in (0,1)$ and $x = N\varepsilon$. Its crossing is ε_N, and $x_N = N\varepsilon_N$. The symbols that only one section uses are defined in that section, where they first appear.

1.2 Main results

Our first theorem gives the center to first order with explicit bounds, and with it the crossing.

Theorem 1.1 (the center and the crossing)**.** *Let $N \geq 2$ be even.*

(a) The ratio C_N/E_N is strictly increasing in $\varepsilon \in (0,1)$, and the crossing ε_N exists, is unique and lies in $(0, \frac{1}{2})$ (Proposition 2.3).

(b) For $0 < \varepsilon < \frac{1}{2}$ we have $C_N \geq \frac{4\varepsilon}{N+2}(1-2\varepsilon)$. If $\varepsilon(1+\log N) \leq \frac{1}{100}$, then $C_N \leq \frac{4\varepsilon}{N+2}\, N^{2\varepsilon} e^{5200\varepsilon}$ (Theorem 2.6, whose constant 5200 uses an enclosure in ball arithmetic).

(c) As $N \to \infty$ over even N, the crossing satisfies $x_N + \log x_N = 2\log N - \log 8 + O(\log^2 N/N)$, and so

$$x_N = 2\log N - \log\log N - \log 16 + O\Big(\frac{\log\log N}{\log N}\Big)$$

(Theorem 2.7).

So $C_N = \frac{4\varepsilon}{N+2}(1 + O(\varepsilon \log N))$ when $\varepsilon \log N$ is small, and the error factor is $1 + O(\log^2 N/N)$ at the crossing. The persistent walk has its crossing at T close to $\log N$, and Theorem S1.7 of Online Resource 1 compares the two crossings to the next order.

For the rest of the law let M_N be the number of steps that disagree with the first step. Its law $f_N(m) = \mathbb{P}(M_N = m)$ does not depend on X_1, and $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)]$ (Lemma 2.2). Let CP_N be the law of $\sum_{d=1}^{N-1} d\,\mathcal{N}_d$, where the $\mathcal{N}_d$ are independent Poisson variables with means $x/(d(d+1))$. Put $\lambda_m = (m - x\log x)/x$, and let

$$f_L(\lambda) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-i\lambda t} e^{B(t)}\, dt, \qquad B(t) = -it\log|t| - \frac{\pi}{2}|t|,$$

be the Landau density.

Theorem 1.2 (the bulk)**.** *(a) For $N \geq 1$ and $0 < \varepsilon < 1$,*

$$\sup_{m \geq 0} \big| f_N(m) - \mathrm{CP}_N(m) \big| \leq \delta_\Sigma := \varepsilon^2(NH_N - 2N + 1) + 2\varepsilon^2(N-1) + 4\varepsilon^2 \sum_{k=0}^{N-2} (H_k^2 + H_k),$$

and $\delta_\Sigma \leq \delta_B := \frac{x^2 \log N}{N} + \frac{4x^2}{N} e^{4x^2/N} + \frac{4x^2(H_N^2 + H_N)}{N}$ (Theorem 3.8).

(b) If $0 < \varepsilon \leq \frac{1}{4}$ and $x \geq 1$, then for every integer $m \geq 0$

$$\big| x f_N(m) - f_L(\lambda_m) \big| \leq \frac{889 + 165\log x}{x} + \frac{6.6\, x}{N} + x\delta_\Sigma \leq 889\Big(\frac{1+\log x}{x} + \frac{x}{N}\Big) + x\delta_B.$$

If $0 < \varepsilon < 1$, $x \to \infty$ and $x^3 \log^2 N/N \to 0$, then $(M_N - x\log x)/x$ converges in law to the law with density f_L (Theorem 3.9).

(c) *The density f_L has exactly one mode λ_{L}, and $-0.22278299 < \lambda_{\mathrm{L}} < -0.22278297$ (Lemma 3.10, computer-assisted). There is K such that for every large N and every ε with $\log N \le x \le 2\log N$, every mode $a^* \ge N/2$ of P_N satisfies*

$$\left|N - a^* - x\log x - \lambda_{\mathrm{L}} x\right| \le K\sqrt{x\log x},$$

and every mode of f_N satisfies the same in place of $N - a^*$ (Corollary 3.11(b)). This holds at the crossing for every large even N. We do not compute K.

So at the crossing the law of A_N has a hump near each end, about $x\log x - 0.2228\,x$ bins from it. The final position is $2A_N - N$, so in physical distance each hump is twice as far from $\pm N$. For the shape of the law between the humps let $D_N(a) = P_N(a+1) - 2P_N(a) + P_N(a-1)$ be its second difference, and for a mode $a^* \ge N/2$ of P_N put $m^* = N - a^*$.

Theorem 1.3 (the shape). *(a) Let $N \ge 3$ and $0 < \varepsilon < \frac{1}{2}$. Delete the zeros from the list $D_N(1), \dots, D_N(N-1)$, replace each entry by its sign, and merge equal neighbors. The result is a subsequence of $(+,-,+,-,+)$. In particular the second difference changes sign at most four times (Theorem 3.18).*

(b) For every even $N \ge 8$, at the crossing, $E_N = C_N < P_N(N-1)$, so P_N is not unimodal (Proposition 3.7, computer-assisted). By contrast f_N is unimodal for every N and every ε (Lemma 3.4, from [7]).

(c) For every large N, even or odd, and every ε with $\log N \le x \le 2\log N$, the following holds. For every mode $a^* \ge N/2$ of P_N we have $D_N(a) > 0$ for $2m^* + 1 \le a \le N - 2m^* - 1$, so P_N is strictly convex on $\{2m^*, \dots, N - 2m^*\}$. This holds at the crossing for every large even N, and there also $P_N(a) > C_N$ for every $1 \le a \le N-1$ with $a \ne h$ (Theorem 3.21 and Corollary 3.23). We do not compute the thresholds. Both conclusions rest on the computer-assisted Lemma 3.10.

(d) At the crossing, the convexity conclusion of (c) holds for every even N with $6 \le N \le 2000$, and the center-minimum conclusion holds for every even N with $8 \le N \le 2000$ (Propositions 3.24 and 3.26, computer-assisted).

For the even N above 2000 and below the thresholds of (c) both conclusions are open (Conjecture 3.14 and Remark 3.27). The second would follow from the first by Lemma 3.22 whenever each mode lies within $(N-2)/4$ of the nearer end.

Three inputs to the proofs rest on computation, and Section 4 describes them. The first is an enclosure in ball arithmetic of a sum of 60 positive terms, which gives the constant 5200 in Theorem 2.6(b). This constant gives the threshold in Proposition 3.7, and in Online Resource 1 the threshold for odd N and the explicit second-order bound. The asymptotic statements need only that the series converges. The second is the enclosure of the mode of the Landau density in Lemma 3.10. It is used in Corollary 3.11, Theorem 3.21, Corollary 3.23 and Proposition S3.1(c) of Online Resource 1. Lean proves this lemma as well, by an integration that its kernel checks, with the unimodality of f_L as its only input. The third is a certified enclosure of ε_N and of the law at the crossing for $10 \le N \le 2000$, and one of the same kind in Lean for $5 \le N \le 9$. It proves Propositions 3.24 and 3.26. For even $N \ge 10$ the enclosure of ε_N also gives Proposition 3.7 for $10 \le N < 1688$. Every other statement is proved by hand, apart from short exact computations in rational arithmetic, and no proof uses the tables or the figures.

Why the elephant needs twice as many reversals. The following argument is heuristic, and Sections 2 and 3 make it precise. The ends are easy. The walk ends in the right end bin exactly when its first step goes right and it never changes direction. This has probability $\frac{1}{2}(1-\varepsilon)^{N-1} \approx \frac{1}{2}e^{-x}$ for the elephant walk, and the same with q in place of ε for the persistent walk. So only the middle differs. In the persistent walk the T switches are spread over the whole walk, and each switch moves the final position by about one run length N/T. So the number of right steps spreads over about $N/\sqrt{T}$ bins, and the middle bin has probability of order $\sqrt{T}/N$. The ends lose when e^{-T} falls to about $1/N$, that is, when $T \approx \log N$.

In the elephant walk a reversal spreads differently. A reversal at step $t+1$ is copied by later steps, those copies are copied, and so on. The steps that descend from it form a family. Its share of the walk behaves like the share of one ball in Pólya's urn started with one ball against t, so a late reversal has a small family and hardly moves the walk. To end exactly in the middle, the walk usually needs a family that fills about half the walk. To first order in ε, a walk that ends in the middle reverses once, among the first few steps. (A walk at the crossing reverses about $x \approx 2\log N$ times in all, but most of these reversals are late and have small families.) An early reversal has probability of order ε, and hitting the exact middle costs another factor of order $1/N$. So the middle bin has probability of order $\varepsilon/N = x/N^2$ (in fact about $4x/N^2$). The ends lose only when e^{-x} falls to about $1/N^2$, that is, when $x \approx 2\log N$. More precisely, the crossing of the persistent walk satisfies $\frac{1}{2}e^{-T} \approx \sqrt{2T/\pi}/N$, and squaring this relation turns it into the elephant relation $\frac{1}{2}e^{-x} \approx 4x/N^2$ with $x \approx 2T - \log(2\pi)$.

The small families also give the shape of the law. About x/m^2 reversals have a family of m steps, so the bin at distance m from an end has probability about $x/(2m^2)$ over a wide range of m. So the law has a hump near each end, and the middle is where the two tails meet. Indeed, twice $x/(2m^2)$ at $m = N/2$ is the $4x/N^2$ above.

1.3 Prior work

The asymptotic behavior of the elephant walk for a fixed memory parameter is well understood [4, 5, 6]. The coin-turning walk of Engländer and Volkov [13] is the persistent walk with a switch probability p_n that depends on the time n. When the switch probability is eventually a/n, that is, $p_n = a/n$ for $n \geq n_0$, the frequency of heads tends to the Beta(a, a) law [13, Section 4, Theorem 1], which is U-shaped for $a < 1$ and humped for $a > 1$, so in that limit the ends and the middle trade places at $a = 1$. Our question is a finite-N version of this comparison for the elephant walk, whose reversal probability depends on N and not on the time.

For $\varepsilon \leq \frac{1}{2}$ the steps of the elephant walk split into clusters, since a step either copies an earlier step exactly, with probability $1-2\varepsilon$, or starts a new cluster with a fair sign. This is Kürsten's coupling with bond percolation on random recursive trees [11], and Baur and Bertoin [14] relate the walk to Pólya-type urns. Our regime $N\varepsilon \asymp \log N$ lies in the strongly supercritical regime of Baur [12], who extends the results of Bertoin [15] for retention probability $1 - t/\log n$. Baur's Corollary 1 gives the sizes of the largest clusters other than the root's (a Poisson limit with intensity $a^{-2}\,da$). Heuristically, in our regime this gives about $2x/m^2$ clusters of size m, since a new cluster starts with probability 2ε. Half of them have the sign opposite to the root, which gives the x/m^2 families of size m behind the tail of the law. These results describe cluster sizes, not the center probability, which is a large deviation of the root cluster. Miyazaki and Takei [16, Theorems 2.3 and 2.4] prove limit theorems for the 'laziest' minimal walk of elephant type at every value of its memory parameter, and with them for the size of the root cluster in percolation on uniform random recursive trees [16, Section 2.3]. The law that appears in the bulk (Theorems 3.8 and 3.9)

is also known. Without its truncation, our compound Poisson law is the Lea–Coulson form of the Luria–Delbrück distribution, and Kessler and Levine [17, Eqs. (14), (18) and (19)] describe its profile $x/(m(m+1))$ for small x and its Landau limit for large x. Möhle [20, Theorem 4.1 and Corollary 4.1] proves this stable limit and gives integral formulas for its density, and his Corollary 4.2 is a local limit theorem for the untruncated law, uniform on the lattice, as the parameter tends to infinity through the integers. It has no rate. Kessler and Levine also note that the law stops being monotone at $\mu N = 2$ and that its maximum sits at about $\mu N(\log \mu N - 0.22)$ for large μN [17, Section 2], where μN plays the role of our x. These are the analogues for the mutant count of the dip at the end (Theorem 3.5) and of the place of the modes (Corollary 3.11). The same stable law governs the fluctuations of the root cluster when the retention probability is $1 - c/\log n$. This follows from a result of Schweinsberg [21], and Bertoin [22] explains it through the growth of the tree. What we add is the truncation at $d = N - 1$, a rate in the local limit, the transfer from this law to the elephant walk in our regime (with an explicit error in Theorem 3.8), the dip and the modes for the walk itself rather than for the compound Poisson law, and a certified enclosure of the Landau mode.

The same model is studied as broadcasting on random recursive trees. Addario-Berry, Devroye, Lugosi and Velona [2] and Althaus, Hartung and Steiner [3] study how well the color of the root can be recovered from the colors at time N, with the flip probability fixed as $N \to \infty$. In that language $N - 2M_N$ is the number of vertices with the color of the root minus the number with the other color, and the center C_N is the probability of a tie, which enters the error of the majority vote. Desmarais, Holmgren and Wagner [23] prove limit laws for the color counts of broadcasting on linear preferential attachment trees, a class that contains the random recursive tree, and Kardas, Shneer and Wachtel [24] give martingale methods and error bounds, again with a fixed flip probability.

Guérin, Laulin, Raschel and Simon [7, Section 2, proof of Theorem 1.1] prove that the law with a fixed first step is unimodal, and their other results fix the memory parameter as N grows. So do the local limit theorem of Trinas and Valle [8, Theorem 3.1] for the superdiffusive walk and their bounds on where log-concavity fails. Gut and Stadtmüller [25, (1.5)] study a walk with steps in $\{0, 1\}$ whose first step is 1 with probability p and whose later steps copy a uniform earlier step with probability p and are 0 otherwise. Given k ones among the first n steps, the next step is 1 with probability pk/n, while for our count A_N the corresponding probability is $\varepsilon + (1 - 2\varepsilon)k/n$. So their count cannot leave 0 and ours can, and their walk does not give our count A_N.

Kessler and Levine [18, Eqs. (10), (13) and (14)] solve the irreversible mutation chain, in which a mutant never reverts, to first order in the mutation probability, with an outer solution that covers numbers of mutants of order N. Its law has the same first derivative in ε at $\varepsilon = 0$ as the law of M_N. We did not find, in the elephant-walk, broadcasting, percolation or Luria–Delbrück literature that we read, a rigorous estimate of the center with a uniform relative error when $N\varepsilon$ is of order $\log N$.

1.4 Organization

Section 2 proves Theorem 1.1, with the bounds on the center proved in Appendix A. It also shows that C_N is a polynomial in ε with $C_N'(0) = 4/(N+2)$ exactly (Proposition 2.5). Section 3 proves Theorems 1.2 and 1.3. The bulk is proved in Appendix C, which uses the bound on the mode of f_N from Appendix B, and the sign pattern of the second difference in Appendix D. Section 3 also bounds the dip $P_N(N-1)/E_N$ at the end from both sides (Theorem 3.5), gives the place of the modes with explicit constants (Corollary 3.11(a)), and shows that one point of convexity

gives convexity between the humps (Theorem 3.19 and Corollary 3.20). Section 4 describes the computations. Online Resource 1 holds the center and the crossing to second order, the comparison with the persistent walk, the tail $P_N(N-m) \sim x/(2m^2)$ beyond the humps, and the crossing and the convexity for odd N. None of Theorems 1.1 to 1.3 uses it.

2 The elephant walk and its crossing

We reduce the elephant walk to a counting chain and prove that the crossing is unique. We then compute the center to first order in ε, bound it from both sides, and locate the crossing.

Definition 2.1 (elephant walk)**.** Fix $0 < \varepsilon < 1$. Let X_1 be uniform on $\{\pm 1\}$. For $t \geq 1$ let U_t be uniform on $\{1, \dots, t\}$, and put $X_{t+1} = X_{U_t}$ with probability $1-\varepsilon$ and $X_{t+1} = -X_{U_t}$ with probability ε. All these choices are independent. We call the event $X_{t+1} = -X_{U_t}$ a flip. As in Section 1.1, $A_N = \#\{i \leq N : X_i = +1\}$ and $P_N(k) = \mathbb{P}(A_N = k)$. The crossing ε_N is the value of ε with $C_N = E_N$ (it exists and is unique by Proposition 2.3), and $x_N = N\varepsilon_N$.

A flip is what the introduction called a reversal. Given the first step, all that matters is how many steps disagree with it. So let M_t be the number of $i \leq t$ with $X_i \neq X_1$, and put

$$r_t(b) = \varepsilon + (1-2\varepsilon)\frac{b}{t}, \qquad f_N(m) = \mathbb{P}_+(M_N = m).$$

Lemma 2.2. *The process $(M_t)_{t \geq 1}$ is a Markov chain with $M_1 = 0$ and $\mathbb{P}(M_{t+1} = b+1 \mid M_t = b) = r_t(b) = 1 - \mathbb{P}(M_{t+1} = b \mid M_t = b)$, and it is independent of X_1. For $0 \leq k \leq N$, $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)]$. In particular $C_N = f_N(h)$, $E_N = \frac{1}{2}(1-\varepsilon)^{N-1}$, and $P_N(k) = P_N(N-k)$.*

Proof. Given $X_1, \dots, X_t$, the copied step X_{U_t} differs from X_1 with probability M_t/t. So $X_{t+1} \neq X_1$ with conditional probability $(1-\varepsilon)M_t/t + \varepsilon(1 - M_t/t) = r_t(M_t)$. This depends only on M_t and not on X_1, which proves the first claim. On $\{X_1 = +1\}$ we have $A_N = N - M_N$, and on $\{X_1 = -1\}$ we have $A_N = M_N$. Averaging over X_1 gives the formula for $P_N(k)$, which does not change under $k \mapsto N-k$. Finally $M_N \leq N-1$, so $f_N(N) = 0$, and $f_N(0) = \prod_{t=1}^{N-1}(1 - r_t(0)) = (1-\varepsilon)^{N-1}$. $\square$

So the end is that of the persistent walk with switch probability ε. More flips always raise the center relative to the end.

Proposition 2.3. *Let $N \geq 2$ and $1 \leq k \leq N-1$. Then $f_N(k)/f_N(0)$ and $P_N(k)/E_N$ are strictly increasing in $\varepsilon \in (0,1)$. In particular C_N/E_N is strictly increasing, the crossing ε_N exists and is unique, and $\varepsilon_N < \frac{1}{2}$. Also $P_N(k) \geq 2E_N$ for $\varepsilon \geq \frac{1}{2}$, and in particular $C_N \geq 2E_N$ there.*

Proof. We compare each path of M with the path that never flips, one step at a time. The probability of a path of M is a product of $N-1$ factors, namely $r_s(b_s)$ for an up step at time s and $1 - r_s(b_s)$ otherwise, where b_s is the state at time s. The path that never flips has probability $f_N(0) = (1-\varepsilon)^{N-1}$, which is one factor $1-\varepsilon$ for each step. So we divide each factor by $1-\varepsilon$. With $y = b_s/s \in [0,1)$ the two kinds of factors become $\frac{\varepsilon + (1-2\varepsilon)y}{1-\varepsilon}$ and $1 - y\frac{1-2\varepsilon}{1-\varepsilon}$, whose derivatives in ε are $\frac{1-y}{(1-\varepsilon)^2} > 0$ and $\frac{y}{(1-\varepsilon)^2} \geq 0$. Both factors are positive, since $r_s(b_s) = \varepsilon(1-y) + (1-\varepsilon)y$ lies strictly between 0 and 1. So the ratio for each path is nondecreasing in ε, and it is strictly increasing if the path has an up step. Summing over the paths with $k \geq 1$ up steps proves the claim for $f_N(k)/f_N(0)$. By Lemma 2.2, $P_N(k)/E_N = [f_N(k) + f_N(N-k)]/f_N(0)$, and both k and $N-k$ are at least 1. So the claim for $P_N(k)/E_N$ follows.

It remains to locate the crossing. The ratio C_N/E_N is continuous and strictly increasing in ε. As $\varepsilon \to 0$ it tends to 0, since $C_N \to 0$ (at $\varepsilon = 0$ the chain never leaves 0) and $E_N \to \frac{1}{2}$. For $\varepsilon \geq \frac{1}{2}$ every $r_s(b)$ and every $1 - r_s(b)$ lies in $[1-\varepsilon, \varepsilon]$, so each path of M has probability at least $(1-\varepsilon)^{N-1} = 2E_N$. Every m with $0 \leq m \leq N-1$ is the end of some path, so $f_N(m) \geq 2E_N$. Both k and $N-k$ lie in $\{1, \dots, N-1\}$, so $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)] \geq 2E_N$, and in particular $C_N \geq 2E_N$ there. So the ratio equals 1 at exactly one ε, and this ε lies in $(0, \frac{1}{2})$. □

2.1 First order, one early flip

When $\varepsilon = 0$ the chain M never leaves 0. After a first flip, and to first order in ε, it is Pólya's urn.

Lemma 2.4 (Pólya's urn)**.** *Let $2 \leq v \leq N$. At time v an urn holds one ball of one color and $v-1$ of the other. At each time $k = v, \dots, N-1$ we draw a ball uniformly and return it with one more ball of its color. Let $\beta_v(d)$ be the probability that at time N there are d balls of the first color. Then $\beta_v(d) = \binom{N-d-1}{v-2} / \binom{N-1}{v-1}$ for $1 \leq d \leq N-v+1$, and $\beta_v(d) = 0$ for other d. Also β_v is nonincreasing on $\{1, \dots, N-v+1\}$ and $\sum_d d\,\beta_v(d) = N/v$. For $1 \leq d \leq N-1$,*

$$\sum_{v=2}^{N} \beta_v(d) = \frac{N}{d(d+1)}, \qquad \sum_{v=2}^{N} (v-2)\,\beta_v(d) = \frac{2N(N-d-1)}{d(d+1)(d+2)}.$$

Proof. We compute the probability of one order of the draws and count the orders. At time k the urn holds k balls. So if s of them are of the first color, the next draw is of that color with probability s/k. Take one order of the $N-v$ draws in which i draws are of the first color. The draws of the first color contribute the numerators $1, 2, \dots, i$, the other draws contribute $v-1, v, \dots, N-i-2$, and the denominators are $v, v+1, \dots, N-1$. So this order has probability $\frac{i!\,(v-1)v\cdots(N-i-2)}{v(v+1)\cdots(N-1)} = \frac{(v-1)\,i!\,(N-i-2)!}{(N-1)!}$, whatever the order is. There are $\binom{N-v}{i}$ orders. With $d = i+1$ this gives $\beta_v(d) = a_d(v-1)\binom{N-v}{d-1}$ with $a_d = \frac{(d-1)!(N-d-1)!}{(N-1)!}$, and writing out the factorials shows that this is $\binom{N-d-1}{v-2}/\binom{N-1}{v-1}$. From the last form, $\beta_v(d)/\beta_v(d-1) = 1 - \frac{v-2}{N-d} \leq 1$ for $2 \leq d \leq N-v+1$, so β_v is nonincreasing on its support. The fraction of balls of the first color is a martingale that starts at $1/v$. So the mean number of such balls at time N is N/v.

For the two sums we use the first form of $\beta_v(d)$ and put $i = N-v$, so that $v-1 = \binom{N-1-i}{1}$ and $(v-1)(v-2) = 2\binom{N-1-i}{2}$. The Chu–Vandermonde identity says that $\sum_i \binom{i}{a}\binom{N-1-i}{b} = \binom{N}{a+b+1}$ (sort the subsets of $\{1, \dots, N\}$ of size $a+b+1$ by their element number $a+1$, which we call $i+1$). With $a = d-1$ and $b = 1, 2$ it shows that the two sums are $a_d\binom{N}{d+1}$ and $2a_d\binom{N}{d+2}$, and these are the values in the statement. □

We need some more notation. Let $V_s(b)$ be the chance of ending at the center from state b at time s. We define it by the backward recursion $V_N(b) = \mathbf{1}\{b = h\}$ and $V_s(b) = r_s(b)V_{s+1}(b+1) + (1 - r_s(b))V_{s+1}(b)$ for $s < N$. So $V_s(b) = \mathbb{P}_+(M_N = h \mid M_s = b)$ whenever $\mathbb{P}_+(M_s = b) > 0$, and $V_s(b)$ is a polynomial in ε. In particular it is defined at $\varepsilon = 0$, without conditioning on an event of probability zero. For $1 \leq t \leq h$ let

$$\pi_t = \beta_{t+1}(h) = \frac{t}{N-t} \prod_{i=1}^{t-1} \frac{h-i}{N-i}, \qquad c_1 = \sum_{t=1}^{h} \pi_t, \qquad w_t = \frac{\pi_t}{c_1},$$

so that $(w_t)_{1 \leq t \leq h}$ is a probability vector.

The first-order coefficient in part (b) below is known. With both switching probabilities equal to ε and one sensitive cell at the start, the number of resistant cells in the reversible fluctuation

assay of Bokes, Hlubinová and Singh [19, Section 2 and Eq. (1)] is our chain M. They condition on the first switch and use Pólya's urn, as we do, and their first-order formula [19, Eqs. (6)–(10)] gives $\varepsilon N/(m(m+1))$ as the part of $f_N(m)$ of first order in ε, which is $\frac{4\varepsilon}{N+2}$ at $m=h$. We include the short proof because the decomposition (a) and the weights (c) are what the uniform bounds of Theorem 2.6 and Appendix A start from.

Proposition 2.5 (first order, as in [19, Eqs. (6)–(10)])**.** *Let $N \geq 2$ be even.*

(a) The center is $C_N = \sum_{t=1}^{h} \varepsilon(1-\varepsilon)^{t-1} V_{t+1}(1)$.

(b) The center C_N is a polynomial in ε with $C_N(0) = 0$ (extended as a polynomial to $\varepsilon = 0$). At $\varepsilon = 0$ we have $V_{t+1}(1) = \pi_t$, and $C_N'(0) = c_1 = \frac{4}{N+2}$ exactly.

(c) The weights satisfy $\sum_t w_t t = 3 - \frac{6}{h+2}$, and $w_t \leq 2t2^{-t}$ for $1 \leq t \leq h$.

Proof. (a) We condition on the time of the first flip. Under $\mathbb{P}_+$ the first flip happens at step $t+1$ exactly when $M_s = 0$ for $s \leq t$ and $M_{t+1} = 1$. Since $r_s(0) = \varepsilon$, this has probability $(1-\varepsilon)^{t-1}\varepsilon$, and by the Markov property the chain then ends at h with probability $V_{t+1}(1)$. A path with no flip ends at $0 \neq h$. So C_N is the sum of $\varepsilon(1-\varepsilon)^{t-1}V_{t+1}(1)$ over $1 \leq t \leq N-1$. Only $t \leq h$ contribute. Indeed at time $t+1$ there are t steps equal to X_1, this number never decreases, and at the center it is h. So $V_{t+1}(1) = 0$ for $t > h$.

(b) Every transition probability is affine in ε, so C_N and $V_{t+1}(1)$ are polynomials in ε. By (a), $C_N(0) = 0$ and $C_N'(0) = \sum_t V_{t+1}(1)|_{\varepsilon=0}$. At $\varepsilon = 0$ we have $r_s(b) = b/s$. So from time $t+1$ on, M counts the balls of the first color in the urn of Lemma 2.4 with $v = t+1$, and $V_{t+1}(1) = \beta_{t+1}(h) = \pi_t$. Since $\beta_v(h) = 0$ for $v > h+1$, Lemma 2.4 with $d = h$ gives $c_1 = \sum_{v=2}^{N} \beta_v(h) = \frac{N}{h(h+1)} = \frac{4}{N+2}$.

(c) In the same way the second sum of Lemma 2.4, with $d = h$ and $v = t+1$, gives $\sum_t (t-1)\pi_t = \frac{2N(h-1)}{h(h+1)(h+2)}$. Dividing by $c_1 = \frac{N}{h(h+1)}$ gives $\sum_t w_t(t-1) = \frac{2(h-1)}{h+2} = 2 - \frac{6}{h+2}$, and adding $\sum_t w_t = 1$ gives the first claim. For the second claim, each factor $\frac{h-i}{N-i}$ in π_t is at most $\frac{1}{2}$, so $\pi_t \leq \frac{t}{N-t}2^{1-t}$. Hence $w_t = \frac{N+2}{4}\pi_t \leq \frac{N+2}{2(N-t)}\, t2^{-t}$. For $t \leq h$ we have $N - t \geq h$ and $\frac{N+2}{2h} = 1 + \frac{2}{N} \leq 2$, so $w_t \leq 2t2^{-t}$. □

So to first order the walk flips once, at some step $t+1$, and then runs as Pólya's urn from t balls of one color and one of the other. As $N \to \infty$ the final fraction of the color that started with one ball tends in law to a Beta$(1,t)$ variable, whose density at $\frac{1}{2}$ is $t2^{1-t}$, and $\sum_t t2^{1-t} = 4$ explains the constant $4/(N+2)$. Since $w_t \leq 2t2^{-t}$, only an early flip counts in this first-order term.

2.2 Two-sided bounds on the center

This subsection bounds the center from both sides. The bounds give the crossing in the next subsection.

Theorem 2.6. *Let $N \geq 2$ be even.*

(a) For $0 < \varepsilon < \frac{1}{2}$, $C_N \geq \frac{4\varepsilon}{N+2}(1-2\varepsilon)$.

(b) The bound $C_N \leq \frac{4\varepsilon}{N+2}\, N^{2\varepsilon} e^{5200\varepsilon}$ holds whenever $0 < \varepsilon(1+\log N) \leq \frac{1}{100}$.

In particular $\log(1-2\varepsilon) \leq \log C_N - \log\frac{4\varepsilon}{N+2} \leq \varepsilon(2\log N + 5200)$ in the range of (b).

Part (a) also holds for $\frac{1}{2} \leq \varepsilon < 1$, trivially, because its right side is not positive there.

Table 1: Elephant crossings x_N and the root g_N of $G(g) = \Lambda_N$. These values are numerical and are not used in any proof. We computed x_N in double precision from the recursion of Lemma 2.2, by Brent's method with an absolute tolerance of 10^{-12} in x, and we found g_N with a tolerance of about 10^{-15}.

N	x_N	g_N	$x_N/\log N$
50	3.940487	4.288636	1.007
400	7.629217	7.843768	1.273
1600	10.240565	10.340051	1.388
3200	11.546834	11.610464	1.431
12800	14.158845	14.182921	1.497
51200	16.778468	16.786946	1.547

Idea of the proof. The proof is in Appendix A. After the first flip, M at time s is Pólya's urn with αs extra balls of each color, where $\alpha = \varepsilon/(1-2\varepsilon)$ (Lemma A.1). They pull the minority fraction toward $\frac{1}{2}$. For (a) we drop them. For (b) we compare M with an urn with $\kappa(s+10)$ extra balls, where $\kappa = 2\varepsilon/(1-4\varepsilon)$. One step of M at time s multiplies the hitting probability of this urn by a factor of at most about $e^{\kappa/s}$, and the product of these factors is about $e^{\kappa H_N} \approx N^{2\varepsilon}$.

2.3 The crossing

By Theorem 2.6, $C_N \approx 4x/N^2$ when $\varepsilon = x/N$ and x is of order $\log N$. Also $E_N = \frac{1}{2}(1-x/N)^{N-1} \approx \frac{1}{2}e^{-x}$. Setting them equal gives $x + \log x \approx 2\log N - \log 8$. So let

$$\Lambda_N = 2L - \log 8, \qquad G(x) = x + \log x,$$

and let g_N be the root of $G(g_N) = \Lambda_N$. It exists and is unique, since G increases from $-\infty$ to ∞ on $(0,\infty)$.

Theorem 2.7. *As $N \to \infty$ over even N, the following hold.*

(a) The crossing satisfies $x_N + \log x_N = \Lambda_N + O(L^2/N)$, so $x_N = g_N + O(L^2/N)$.

(b) The crossing satisfies $x_N = 2\log N - \log\log N - \log 16 + O(\log\log N/\log N)$.

Proof. (a) *Step 1. Both sides near the crossing.* Put $F(x) = \log C_N(x/N) - \log E_N(x/N)$ for $0 < x < N$, where $C_N(\varepsilon)$ and $E_N(\varepsilon)$ are the center and the end at flip probability ε. By Proposition 2.3, F is strictly increasing and x_N is its only zero. We show that

$$|F(x) - G(x) + \Lambda_N| \le \rho_N \quad (1 \le x \le 2\Lambda_N), \qquad \text{with } \rho_N = O(L^2/N). \tag{$*$}$$

Let $1 \le x \le 2\Lambda_N$ and $\varepsilon = x/N$. Then $\varepsilon \le 4L/N$, so for large N both parts of Theorem 2.6 apply, and together they say that $\log C_N = \log \frac{4\varepsilon}{N+2} + O(\varepsilon L)$. Here $\log \frac{4\varepsilon}{N+2} = \log x + \log 4 - 2L - \log(1+2/N)$, and $\log(1+2/N) \le 2/N \le 2xL/N$ because $x \ge 1$. For the end we use $\log(1-y) = -y + O(y^2)$ for $0 \le y \le \frac{1}{2}$ in $\log E_N = -\log 2 + (N-1)\log(1-\frac{x}{N})$. So

$$\log C_N = \log x + \log 4 - 2L + O\Big(\frac{xL}{N}\Big), \qquad \log E_N = -\log 2 - x + O\Big(\frac{x^2}{N}\Big).$$

We subtract. Since $\log 4 + \log 2 = \log 8$, the main terms give $x + \log x - 2L + \log 8 = G(x) - \Lambda_N$, and both errors are $O(L^2/N)$ because $x \le 2\Lambda_N \le 4L$. This is ($*$).

Keeping the constants gives an explicit ρ_N. Suppose that $400L(1+L) \le N$. Then $\varepsilon(1+\log N) \le 4L(1+L)/N \le \frac{1}{100}$, so the inequality after part (b) of Theorem 2.6 applies. With $\log(1-y) \ge -y-y^2$ for $0 \le y \le \frac{1}{2}$ and $x \le 4L$, the error term for $\log C_N$ above lies between $-(3+8L)/N$ and $(8L^2+20800L)/N$, and the one for $\log E_N$ lies between $-16L^2/N$ and $4L/N$. So $(*)$ holds with $\rho_N = (24L^2+20800L)/N$.

Step 2. Trap the zero. We have $G' = 1 + 1/x \ge 1$ and $G(g_N) = \Lambda_N$. So $G(x) - \Lambda_N < -\rho_N$ for $x < g_N - \rho_N$, and $G(x) - \Lambda_N > \rho_N$ for $x > g_N + \rho_N$. By $(*)$, $F(x) < 0$ for $1 \le x < g_N - \rho_N$ and $F(x) > 0$ for $g_N + \rho_N < x \le 2\Lambda_N$. Both ranges are nonempty for large N, because $\Lambda_N - \log\Lambda_N \le g_N \le \Lambda_N$ (note that $G(\Lambda_N - \log\Lambda_N) \le \Lambda_N \le G(\Lambda_N)$). Since F is increasing, its zero x_N lies between the two ranges, so $|x_N - g_N| \le \rho_N$. Then x_N lies in $[1, 2\Lambda_N]$, and $F(x_N) = 0$ in $(*)$ gives $|G(x_N) - \Lambda_N| \le \rho_N$. With the explicit ρ_N above, these two bounds hold for every even N with $L \ge 30$. Indeed $e^L \ge L^{12}/12!$ gives $24L^4 + 20800L^3 \le N$, so $400L(1+L) \le N$ and $\rho_N \le 1/L^2$, and both ranges are nonempty because $g_N \ge \Lambda_N - \log\Lambda_N > 50$.

(b) We solve $g_N = \Lambda_N - \log g_N$ by one substitution. Since $\Lambda_N - \log\Lambda_N \le g_N \le \Lambda_N$, the ratio g_N/Λ_N is $1 + O(\log\Lambda_N/\Lambda_N)$, so $\log g_N = \log\Lambda_N + O(\log\Lambda_N/\Lambda_N)$. Also $\Lambda_N = 2\log N - \log 8$ gives $\log\Lambda_N = \log 2 + \log\log N + O(1/\log N)$. So $g_N = 2\log N - \log 8 - \log 2 - \log\log N + O(\log\log N/\log N)$, which is (b) for g_N, and (a) gives it for x_N. □

Table 1 gives crossings for six values of N of the form $50 \cdot 2^k$. The limit 2 of $x_N/\log N$ is approached slowly, because of the term $-\log\log N$.

In Online Resource 1, Proposition S1.1 proves a lower bound for C_N that matches Theorem 2.6(b) up to a factor $1 + O(\varepsilon)$, and Proposition S1.2 gives one more term of the crossing, which shows that $x_N < g_N$ for large N. Theorem S1.7 there compares x_N with the crossing value z_N of T for the persistent walk. It shows that $x_N - (2z_N - \log(2\pi))$ is $c_*/\log N$ to first order, with $c_* = \frac{1}{2}\log(2\pi) - \frac{1}{4}$.

3 The elephant walk near its crossing

We now look at the whole law P_N near the crossing. In this section and in Appendices B and C, a statement that says "Assume (H)" assumes

$$N \ge 64, \qquad 0 < \varepsilon \le \tfrac{1}{4}, \qquad x = N\varepsilon \ge 1. \tag{H}$$

Every other statement holds under its own hypotheses. Recall that $P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)]$. Some statements also use odd N. There x_N is the value of x at which the two central bins are as likely as the ends (it exists and is unique by Proposition 3.25). For a mode $a^* \ge N/2$ of P_N we put $m^* = N - a^*$, and m_h is the largest mode of f_N.

3.1 Families and the first flip

For $k_0 \ge 1$ let $(M_k^{[k_0]})_{k \ge k_0}$ be the Markov chain with $M_{k_0}^{[k_0]} = 0$ and the transition probabilities $r_k(b)$ of M. It is the count of disagreeing steps for a walk whose first k_0 steps are forced to agree. So $M^{[1]}$ has the law of M. We write $Y_d = M_d^{[1]}$, and $Z_t^{(j)} = M_t^{[j-1]}$ for $j \ge 2$ and $t \ge j-1$. In a random recursive tree on $[N] = \{1, \dots, N\}$, vertex $k+1$ is joined to a uniform vertex of $[k]$, independently for $k = 1, \dots, N-1$. We write $\mathcal{T}_v$ for the subtree of v, that is, v and its descendants, and we call it the family of v.

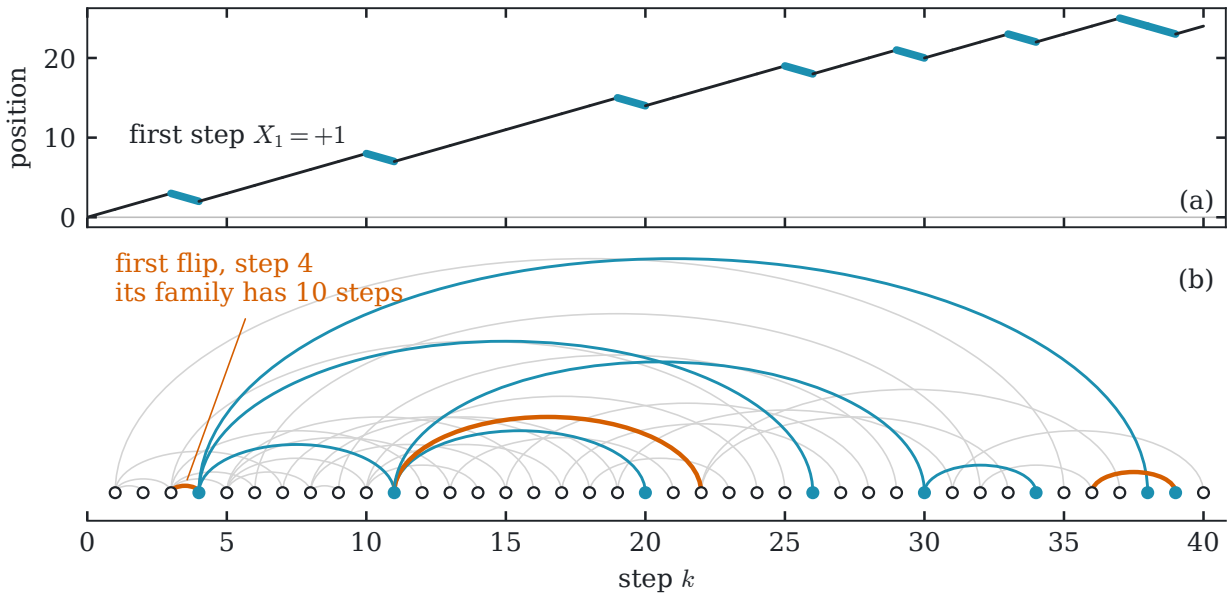


Figure 1: One elephant walk with $N = 40$ steps and $\varepsilon = 0.07$. (a) The position against the step. The steps that disagree with the first step are drawn heavier and in blue. (b) The memory tree of the same walk. Step $k+1$ copies the step at the left end of its arc, and the arcs drawn heavier are flips. A step disagrees with the first step exactly when the path from it to step 1 crosses an odd number of flips (Lemma 3.1). The first flip, at step 4, turns its family of ten steps. The flip at step 22, inside that family, turns its own family of three steps back, and the flip at step 39 turns only that step.

Lemma 3.1 (tree and marks)**.** *Let $\mathcal{T}$ be the tree on $[N]$ in which the parent of $k+1$ is U_k. For $2 \le v \le N$ let $\mu_v = 1$ if step v flips the step it copies, and $\mu_v = 0$ otherwise. Then $\mathcal{T}$ is a random recursive tree, the marks μ_v are i.i.d. Bernoulli(ε) and independent of $\mathcal{T}$, and $X_v = X_1(-1)^{\sigma(v)}$, where $\sigma(v)$ is the number of marks on the path from v to the root, the root excluded. So $M_N = \#\{v : \sigma(v) \text{ odd}\}$. If the marks on $2, \dots, k_0$ are set to 0, then the number of odd vertices in $[k]$, $k \ge k_0$, is a copy of the chain $M^{[k_0]}$.*

Proof. The parent U_k of $k+1$ is uniform on $[k]$ and independent of the earlier choices, and the marks are the flip indicators. The path from $k+1$ to the root is $k+1$ followed by the path from U_k, so $\sigma(k+1) = \mu_{k+1} + \sigma(U_k)$ and, by induction on k, $X_{k+1} = X_{U_k}(-1)^{\mu_{k+1}} = X_1(-1)^{\sigma(k+1)}$. Now set $\mu_2 = \cdots = \mu_{k_0} = 0$. If b of the vertices in $[k]$ are odd, $k \ge k_0$, then $k+1$ is odd with conditional probability $(1-\varepsilon)\frac{b}{k} + \varepsilon(1 - \frac{b}{k}) = r_k(b)$, whatever else is known about $[k]$. □

So a single flip at v turns every step of its family (Figure 1). For $k \ge v$, if $\mathcal{T}_v$ contains s of the vertices in $[k]$, then $k+1$ joins $\mathcal{T}_v$ with probability s/k. This is the urn of Lemma 2.4, so $|\mathcal{T}_v|$ has the law β_v and mean N/v, and on average there are $N/(d(d+1))$ families of size d. If v is the only marked vertex, the odd vertices are those of $\mathcal{T}_v$. Two or more marks have probability $O(\varepsilon^2)$ at fixed N, and no mark gives $M_N = 0$. So at fixed N the part of $f_N(m)$ of first order in ε is exactly $\varepsilon \sum_v \beta_v(m) = x/(m(m+1))$ for $1 \le m \le N-1$, which is the formula of [19, Eq. (10)] with one cell at the start. It is also proved in Lean for the chain M (Section 4).

To go beyond first order we cut the tree at the first flip.

Lemma 3.2 (splitting)**.** *Let $\mathcal{T}$ be a random recursive tree on $[N]$, let $2 \le v \le N$ and let $S \subseteq [N]$ with $\min S = v$. Given that $\mathcal{T}_v$ has vertex set S, the restrictions of $\mathcal{T}$ to S and to $[N] \setminus S$, relabeled in increasing order, are independent random recursive trees. They are independent of the parent of v, which is uniform on $[v-1]$.*

Proof. The tree $\mathcal{T}$ is uniform over the $(N-1)!$ increasing trees on $[N]$. Its family $\mathcal{T}_v$ has vertex set S exactly when every $u \in S \setminus \{v\}$ has its parent in S and every $u \notin S$ with $u \ge 2$ has its parent outside S. So, given this event, each element of S other than v picks its parent freely among the smaller elements of S, v picks any element of $[v-1]$, and each element of $[N] \setminus S$ other than 1 picks freely among the smaller elements of $[N] \setminus S$. This is the claim. □

Lemma 3.3 (the first flip)**.** *Let $V = \min\{v \ge 2 : \mu_v = 1\}$, with $V = \infty$ and $|\mathcal{T}_V| = 0$ if there is no mark. If $V < \infty$, let $\bar{Y}$ be the number of vertices u of $\mathcal{T}_V$ with $\sigma(u)$ even, and let $\bar{Z}$ be the number of odd vertices outside $\mathcal{T}_V$. Then for $2 \le j \le N$, $1 \le d \le N-j+1$ and integers y, z,*

$$\mathbb{P}\big(V = j,\, |\mathcal{T}_V| = d,\, \bar{Y} = y,\, \bar{Z} = z\big) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)\,\mathbb{P}(Y_d = y)\,\mathbb{P}\big(Z^{(j)}_{N-d} = z\big).$$

In particular $\mathbb{P}(V = j,\, |\mathcal{T}_V| = d) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and for $m \ge 1$

$$f_N(m) = \sum_{j=2}^{N} \varepsilon(1-\varepsilon)^{j-2} \sum_{d=1}^{N-j+1} \beta_j(d)\,\mathbb{P}\big(d - Y_d + Z^{(j)}_{N-d} = m\big),$$

where Y_d and $Z^{(j)}_{N-d}$ are independent.

Proof. Marks and tree are independent, and $|\mathcal{T}_j|$ has the law β_j, which gives the law of $(V, |\mathcal{T}_V|)$. Condition on $V = j$ and on the vertex set S of $\mathcal{T}_j$, with $|S| = d$. The marks on $j+1, \dots, N$ are still i.i.d., and by Lemma 3.2 the trees on S and on $[N] \setminus S$ are independent random recursive trees. The vertices $1, \dots, j-1$ carry no mark, so j is odd, and $u \in S$ is even exactly when the path from u up to j, with j excluded, has an odd number of marks. So by Lemma 3.1 for the tree on S rooted at j, the count $\bar{Y}$ has the law of Y_d. Paths from outside S stay outside S, and the tree on $[N] \setminus S$ has the unmarked vertices $1, \dots, j-1$ first. So by Lemma 3.1 the count $\bar{Z}$ has the law of $Z^{(j)}_{N-d}$. The two counts are functions of the two trees and of the marks on the two disjoint sets S and $[N] \setminus S$, so they are independent, and their joint law depends on S only through d. This gives the joint law. On $\{V = j\}$ we have $M_N = d - \bar{Y} + \bar{Z}$, and $M_N = 0$ if there is no mark. Summing over j, d and the pairs (y, z) with $d - y + z = m$ gives the formula for $f_N(m)$. □

So the first flip turns its whole family except the steps of that family that flip back, and the rest of the tree is a walk whose first $j-1$ steps agree with X_1. Only the law of each pair $(Y_d, Z^{(j)}_{N-d})$ enters the lemma. So from now on $(Y_d)_{d \ge 1}$ is a copy of the chain $M^{[1]}$, and $(Z^{(j)}_t)_{t \ge j-1}$ is a copy of $M^{[j-1]}$, independent of Y and of $(V, |\mathcal{T}_V|)$.

Lemma 3.4 (unimodality given the first step [7, Section 2, proof of Theorem 1.1])**.** *The function f_N is unimodal.*

This is shown at the start of the proof of Theorem 1.1 of [7], for every N and every memory parameter $p = 1 - \varepsilon \in [0, 1]$, for the position $N - 2M_N$ given $X_1 = +1$. The lemma is also proved in Lean (Section 4). So f_N is nondecreasing on $\{0, \dots, m_h\}$ and nonincreasing on $\{m_h, \dots, N\}$. With a fair first step, P_N mixes f_N and its mirror image. When the mode of f_N is far from the center, the mixture can have two humps, one near each end.

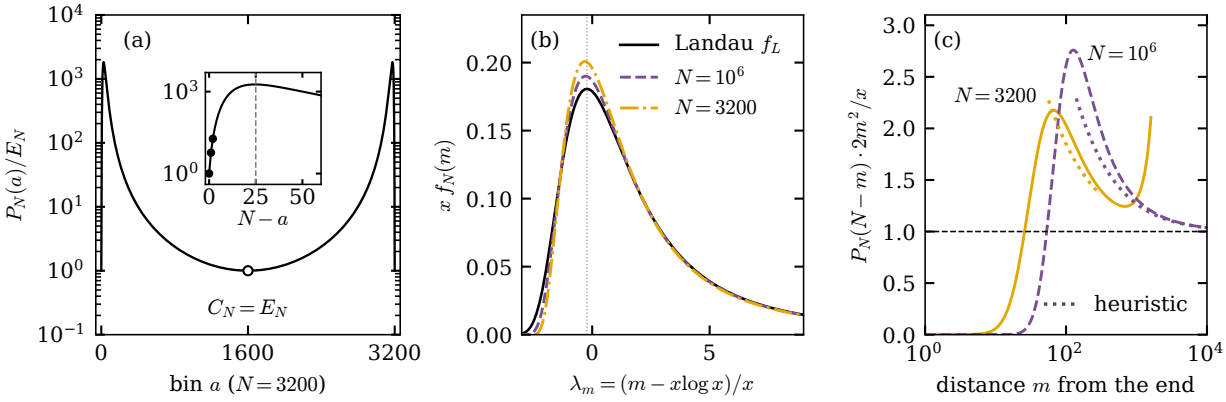


Figure 2: The elephant walk at its crossing. (a) The law P_N for $N = 3200$ at its crossing $x = 11.5468$, divided by the end E_N. The center equals the end, and the law rises from each end to a mode and falls to the center. The inset shows the last 60 bins, with dots at bins N, $N-1$ and $N-2$. The end is below its neighbor, and the modes are 25 bins from each end (dashed). (b) The law of M_N, scaled as $xf_N(m)$, against $\lambda_m = (m - x\log x)/x$, with the Landau density f_L. The dotted line is its mode λ_{L}. (c) The tail $P_N(N-m)$ divided by $x/(2m^2)$. The dashed line is the limit 1 of Theorem S2.1 of Online Resource 1, and the dotted curves are the heuristic correction $1 + x(2\log m + 2\gamma - 3)/m$ for $2x\log x \le m \le N/x$. The rise of the $N = 3200$ curve near $m = N/2$ is the mirror term $f_N(N-m)$. At $N = 10^6$ we use $x = g_N = 22.4407$.

3.2 The end and its neighbor

We first say heuristically why the end should fall below its neighbor once $x > 2$ (Figure 2(a), inset). Given $X_1 = +1$, the end needs no flip at all. Its neighbor, bin $N-1$, needs one flip that is never copied. A flip at step j is never copied with probability about j/N. Summing $\varepsilon j/N$ over j gives a factor of about $x/2$ in favor of the neighbor.

Theorem 3.5 (endpoint dip)**.** *Let $N \ge 2$ and $0 < \varepsilon < 1$, and put $\rho_1 = P_N(N-1)/E_N$. Then*

$$\rho_1 > \frac{x}{2(1-\varepsilon)}.$$

If moreover $N \ge 3$ and $\varepsilon \le \frac{1}{2}$, then

$$\rho_1 \le \frac{x(1+1/N)^2}{2-3\varepsilon} + \frac{\varepsilon}{1-\varepsilon}.$$

Hence the end is strictly below its neighbor if $x \ge 2(1-\varepsilon)$. If $N \ge 3$, $\varepsilon \le \frac{1}{2}$ and the right side of the second bound is below 1, the end is strictly above its neighbor.

Proof. By Lemma 2.2, $\rho_1 = \big(f_N(1) + f_N(N-1)\big)/f_N(0)$ with $f_N(0) = (1-\varepsilon)^{N-1}$. We bound the two terms in turn.

The term $f_N(1)$. On the event $M_N = 1$ the chain moves up exactly once, say from time $j-1$ to time j with $2 \le j \le N$. Before that it stays at 0 for $j-2$ steps, each with probability $1 - r_t(0) = 1 - \varepsilon$. The move up has probability ε. After it the chain stays at 1 at the times $k = j, \dots, N-1$, each with probability $1 - r_k(1) = (1-\varepsilon)(1-c'/k)$, where $c' = (1-2\varepsilon)/(1-\varepsilon) \le 1$.

Multiplying and summing over j gives

$$f_N(1) = \varepsilon(1-\varepsilon)^{N-2} \sum_{j=2}^{N} \vartheta_j, \qquad \vartheta_j = \prod_{k=j}^{N-1} \Big(1 - \frac{c'}{k}\Big).$$

For the lower bound we use $c' \le 1$. Then $\vartheta_j \ge \prod_{k=j}^{N-1}(1-1/k) = (j-1)/(N-1)$, and these lower bounds sum to $N/2$ over $2 \le j \le N$. So $f_N(1)/f_N(0) \ge x/(2(1-\varepsilon))$. For the upper bound let $N \ge 3$ and $\varepsilon \le \frac{1}{2}$, so that $c' \ge 0$. Then $1 - c'/k \le e^{-c'/k}$ and $\sum_{k=j}^{N-1} 1/k \ge \log(N/j)$ give $\vartheta_j \le (j/N)^{c'}$. As $(t/N)^{c'}$ is nondecreasing in t, we can compare the sum with an integral, namely

$$\sum_{j=2}^{N} \vartheta_j \le \int_2^{N+1} \Big(\frac{t}{N}\Big)^{c'} dt \le \frac{N(1+1/N)^{c'+1}}{c'+1} \le \frac{N(1+1/N)^2(1-\varepsilon)}{2-3\varepsilon},$$

using $c' \le 1$ and $1/(c'+1) = (1-\varepsilon)/(2-3\varepsilon)$. So $f_N(1)/f_N(0) \le x(1+1/N)^2/(2-3\varepsilon)$.

The term $f_N(N-1)$. The event $M_N = N-1$ needs an up move at every time. The first has probability $r_1(0) = \varepsilon$, and the one at time $k \ge 2$ has probability $r_k(k-1) = 1 - \varepsilon - (1-2\varepsilon)/k$. This lies strictly between 0 and 1, as every $r_k(b)$ with $0 < b < k$ does, and it lies in $[\frac{1}{2}, 1-\varepsilon]$ when $\varepsilon \le \frac{1}{2}$. So $f_N(N-1) > 0$, and $f_N(N-1) \le \varepsilon(1-\varepsilon)^{N-2}$ when $\varepsilon \le \frac{1}{2}$. This makes the lower bound strict and gives the term $\varepsilon/(1-\varepsilon)$ in the upper bound. (For $N = 2$ the bins 1 and $N-1$ coincide, and the argument is the same.) □

Corollary 3.6 (the dip at the crossing)**.** *Let $N \ge 2$ be even and $0 < \varepsilon < 1$, and suppose that $x \ge 2(1-\varepsilon)$ and $C_N \le E_N$. Then $C_N \le E_N < P_N(N-1)$, and P_N is not unimodal. In particular, if $x_N \ge 2(1-\varepsilon_N)$, then at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal.*

Proof. Theorem 3.5 gives $P_N(N-1) > E_N \ge C_N$. Suppose that P_N is nondecreasing on $\{0, \dots, a_0\}$ and nonincreasing on $\{a_0, \dots, N\}$. It is symmetric about h, so we may replace a_0 by $N - a_0$ and assume $a_0 \le h$. Then $P_N(a) \le C_N$ for every $a \ge h$, including $a = N-1$, which is a contradiction. □

By Proposition 2.3, $C_N \le E_N$ holds exactly when $\varepsilon \le \varepsilon_N$, and $x \ge 2(1-\varepsilon)$ says that $\varepsilon \ge 2/(N+2)$. So the corollary covers the ε with $2/(N+2) \le \varepsilon \le \varepsilon_N$. In particular the condition $x_N \ge 2(1-\varepsilon_N)$ says that $\varepsilon_N \ge 2/(N+2)$. By Proposition 2.3 it holds exactly when $C_N \le E_N$ at $\varepsilon = 2/(N+2)$. We now show that this is so for every even $N \ge 8$.

Proposition 3.7 (the hypothesis holds from $N = 8$ on, computer-assisted)**.** *For every even $N \ge 8$ we have $C_N < E_N$ at $\varepsilon = 2/(N+2)$, and hence $x_N > 2(1-\varepsilon_N)$. So for every even $N \ge 8$, at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal. For $N = 2$, 4 and 6 the condition $x_N \ge 2(1-\varepsilon_N)$ fails, and no ε meets the hypotheses of Corollary 3.6.*

Proof. The range $N \ge 1688$. Take $\varepsilon = 2/N$. Then $\varepsilon(1+\log N) \le \frac{1}{100}$, so Theorem 2.6(b) gives $C_N \le \frac{8}{N(N+2)} N^{4/N} e^{10400/N}$. The right side decreases in N and is below $1.4 \cdot 10^{-3}$ at $N = 1688$. Since $\log(1-y) \ge -y - y^2$ for $0 \le y \le \frac{1}{2}$, we also have $E_N = \frac{1}{2}(1 - \frac{2}{N})^{N-1} \ge \frac{1}{2} e^{-2-2/N} > 0.06$. So $C_N < E_N$ at $\varepsilon = 2/N$, and Proposition 2.3 gives $\varepsilon_N > 2/N$. Hence $C_N < E_N$ at $\varepsilon = 2/(N+2)$ as well.

The range $8 \le N < 1688$. For $N = 8$ the ratio C_N/E_N at $\varepsilon = 2/(N+2)$ is exactly $\frac{67903}{71680} < 1$. For even $10 \le N \le 2000$ the script `certify_center.py` encloses ε_N in ball arithmetic as in the proof of Proposition 3.24, and it checks by an exact rational comparison that the lower end of

the enclosure lies above $2/(N+2)$. So $\varepsilon_N > 2/(N+2)$, and Proposition 2.3 gives $C_N < E_N$ at $\varepsilon = 2/(N+2)$. The ratio of the lower end to $2/(N+2)$ is at least 1.107, with its minimum at $N = 10$.

The three smallest N. At $\varepsilon = 2/(N+2)$ the ratio C_N/E_N is exactly 2, $\frac{5}{3}$ and $\frac{508}{405}$ for $N = 2$, 4 and 6, so $\varepsilon_N < 2/(N+2)$ there. These ratios and the one for $N = 8$ are computed in exact rational arithmetic by `certify_elephant.py`. The last claim of the proposition about $N \geq 8$ is Corollary 3.6. □

3.3 The bulk

The steps that disagree with X_1 lie in the flipped families. Recall the compound Poisson law CP_N of Section 1.2. On average $x/(d(d+1))$ flipped vertices have a family of size d. If no flipped step has a flipped descendant, which is likely when $x^2 \log N/N$ is small, then M_N is the total size of the flipped families, and this is close to the compound Poisson sum.

The law CP_N and its Landau limit are known (Section 1.3). What the two theorems below add is the truncation at $d = N-1$, a rate with explicit constants in both theorems, and the transfer to the elephant walk in sup norm. Numerical values of f_L and of its mode are standard (see Kölbig and Schorr [26]). What Lemma 3.10 adds is a certified enclosure.

Theorem 3.8 (compound Poisson bulk)**.** *For $N \geq 1$ and $0 < \varepsilon < 1$,*

$$\sup_{m \geq 0} \bigl| f_N(m) - \mathrm{CP}_N(m) \bigr| \leq \delta_\Sigma := \varepsilon^2 (N H_N - 2N + 1) + 2\varepsilon^2 (N-1) + 4\varepsilon^2 \sum_{k=0}^{N-2} (H_k^2 + H_k).$$

Also $\delta_\Sigma \leq \delta_B := \frac{x^2 \log N}{N} + \frac{4x^2}{N} e^{4x^2/N} + \frac{4x^2 (H_N^2 + H_N)}{N}$.

Bin by bin, the law of the number of disagreeing steps is within δ_Σ of the compound Poisson law. Bounding each H_k by H_N gives $\delta_\Sigma \leq \frac{x^2}{N}(\log N + 2 + 4H_N^2 + 4H_N)$, so $\delta_\Sigma = O(x^2 \log^2 N/N)$.

Idea of the proof. The proof is in Appendix C. Let Z_+ be the total size of the flipped families. Then $M_N = Z_+$ unless some flipped step has a flipped descendant, and this has probability at most $x^2 \log N/N$. Given the tree, Z_+ is a sum of independent terms, so its characteristic function is a product, and the product is close to $e^{\varepsilon F}$ with $F = \sum_v (e^{i\theta |\mathcal{T}_v|} - 1)$. The mean of εF is exactly the logarithm of the characteristic function of CP_N. So it remains to show that F is close to its mean. For this we use the Efron–Stein inequality [27]. Changing the parent of one vertex changes the family sizes only along two paths to the root, and these paths have length of order $\log N$.

The families of size up to about x add up to nearly $x \log x$, and the few families of size of order x give fluctuations of order x. So we center M_N at $x \log x$ and divide by x, which gives the coordinate λ_m and the Landau density f_L of Section 1.2. It is a stable density with index one. We show in Appendix C that e^B is the characteristic function of a probability law of class L, so that f_L is a unimodal probability density by [28].

Theorem 3.9 (Landau limit)**.** *(a) Let $x \geq 1$ and $N \geq 4x$. For every integer m, with $\mathrm{CP}_N(m) = 0$ for $m < 0$,*

$$\bigl| x\, \mathrm{CP}_N(m) - f_L(\lambda_m) \bigr| \leq E_0 := \frac{889 + 165 \log x}{x} + \frac{6.6\, x}{N} \leq 889 \Bigl(\frac{1 + \log x}{x} + \frac{x}{N} \Bigr).$$

(b) Let $0 < \varepsilon \le \frac{1}{4}$ and $x \ge 1$. For every integer $m \ge 0$,

$$\bigl|x f_N(m) - f_L(\lambda_m)\bigr| \le E_0 + x\delta_\Sigma \le 889\Bigl(\frac{1+\log x}{x} + \frac{x}{N}\Bigr) + x\delta_B.$$

(c) Let $N_n \ge 1$ and $0 < \varepsilon_n < 1$ for $n \ge 1$, and put $x_n = N_n\varepsilon_n$. If $x_n \to \infty$ and $x_n^3 \log^2 N_n/N_n \to 0$, then $(M_{N_n} - x_n \log x_n)/x_n$ converges in law to the law with density f_L.

Near the crossing the number of disagreeing steps is about $x \log x$, with fluctuations of order x that follow the Landau law, bin by bin (Figure 2(b)). Both conditions of (c) hold at the crossing, by Theorem 2.7.

Idea of the proof. The proof is in Appendix C. If we allow every jump size $d \ge 1$, the characteristic function of the compound Poisson law is $e^{x\chi(e^{i\theta})}$ with $\chi(s) = (1-s)\log(1-s)/s$, and the jumps $d \ge N$ change it little. For small θ we have $\chi(e^{i\theta}) = -i\theta\log(-i\theta) + O(\theta^2|\log|\theta||)$. With $\theta = t/x$ the exponent becomes $B(t) + it\log x$ plus an error of order $t^2 \log x/x$. The term $it\log x$ is the centering at $x\log x$. Both characteristic functions decay like $e^{-c|t|}$, so Fourier inversion turns this into a bound for each bin, and keeping the constants gives 889 and 165. Part (b) then follows from Theorem 3.8.

To locate the modes we need to know that f_L is curved at its maximum.

Lemma 3.10 (the mode of the Landau density, computer-assisted)**.** *(a) The density f_L has exactly one mode λ_{L}, and $-0.22278299 < \lambda_{\mathrm{L}} < -0.22278297$. It is nondecreasing on $(-\infty, \lambda_{\mathrm{L}}]$ and nonincreasing on $[\lambda_{\mathrm{L}}, \infty)$.*

(b) The derivative vanishes at the mode, $f_L'(\lambda_{\mathrm{L}}) = 0$, and $-0.0792 \le f_L''(\lambda) \le -0.0786$ for $-0.2229 \le \lambda \le -0.2227$.

(c) For every real λ, $f_L(\lambda_{\mathrm{L}}) - f_L(\lambda) \ge \kappa_L \min\{1, (\lambda - \lambda_{\mathrm{L}})^2\}$ with $\kappa_L = 2.6 \cdot 10^{-10}$.

The proof is in Appendix C.3, and it is computer-assisted. Lean proves the lemma as well, with the unimodality of f_L that [28] gives as its only input (Section 4), so the computation in ball arithmetic is an independent check.

Corollary 3.11 (where the modes are)**.** *(a) Assume (H). Let κ_L be as in Lemma 3.10(c), let $E_1 = 889\bigl(\frac{1+\log x}{x} + \frac{x}{N}\bigr) + x\delta_B$ be the bound of Theorem 3.9(b), and put*

$$E' = E_1 + \frac{16x^2 H_N}{N^2} + \frac{\|f_L''\|_\infty}{8x^2}.$$

If $E' < \kappa_L/2$ and $m_h \le N/4$, then every mode $a^* \ge N/2$ of P_N satisfies $|m^* - x\log x - \lambda_{\mathrm{L}} x| \le x\sqrt{2E'/\kappa_L}$, and so does every mode of f_N.

(b) There are K and N_2 such that the following holds for every $N \ge N_2$, even or odd, and every ε with $\log N \le x \le 2\log N$. Every mode $a^* \ge N/2$ of P_N has

$$\bigl|m^* - x\log x - \lambda_{\mathrm{L}} x\bigr| \le K\sqrt{x\log x},$$

and so has every mode of f_N in place of m^*. In particular this holds at the crossing, with $x = x_N$, for every large even N (Theorem 2.7(b)).

In (a) every constant is explicit. (Here $\|f_L''\|_\infty \le 16/\pi^4$, since $|e^{B(t)}| = e^{-\pi|t|/2}$.) We do not compute K and N_2 in (b).

So each hump of the law at the crossing sits about $x \log x - 0.2228\, x$ bins from its end. The proof is in Appendix C.4. It compares f_N at a mode with f_N at the integer nearest to $x \log x + \lambda_{\mathrm{L}} x$, and uses Theorem 3.9(b) and Lemma 3.10(c).

Remark 3.12 (how good the bulk approximation is). The bounds of this subsection are far from sharp at the sizes we can compute. At $N = 10^6$ and $x = 22.4407$ the bound δ_B is still about 0.46, but there the computed law has $|f_N(m)/\mathrm{CP}_N(m) - 1| \le 3.6 \cdot 10^{-4}$ for $1 \le m \le 10^4$. At every tested crossing m^* equals the mode of CP_N. The proof of Corollary 3.11(b) needs the bound of Theorem 3.9(b) to be below $\kappa_L/2$, where $\kappa_L = 2.6 \cdot 10^{-10}$ is the constant of Lemma 3.10(c). With the constant 889 this needs $x > 2 \cdot 10^{14}$, and numerically the best κ_L is about 0.026. These numbers are computed in double precision and are not used in any proof.

3.4 The tail

Beyond the humps the law falls like $x/(2m^2)$. Theorem S2.1 of Online Resource 1 proves that $P_N(N-m) = \frac{x}{2m^2}(1 + o(1))$ for even N with $x \ge 1$ and $x \log^2 N/N \to 0$, uniformly for $\omega_N x \log^2(x+1) \le m \le \delta'_N N$, for every $\omega_N \to \infty$ and every $\delta'_N \to 0$.

At $N = 10^6$, $x = 22.4407$ and $m = 10^4$ the computed ratio $P_N(N-m)/(x/(2m^2))$ is 1.03777. Heuristically, at moderate m the law CP_N suggests $f_N(m)\, m(m+1)/x \approx 1 + x(2\log m + 2\gamma - 3)/m$, which we do not prove. This factor is still about 2 near $m = 70$ when $N = 3200$ (Figure 2(c)).

3.5 Convexity between the humps

Definition 3.13 (convex between the humps)**.** Let $N \ge 5$. The law at the crossing is *convex between the humps* if, for $x = x_N$ and every mode $a^* \ge N/2$ of P_N with $m^* = N - a^*$, the second difference $P_N(a-1) - 2P_N(a) + P_N(a+1)$ is positive for $2m^* + 1 \le a \le N - 2m^* - 1$.

With the symmetry of the law, this makes P_N strictly convex on $[2m^*, N - 2m^*]$, so h is its unique minimizer there (the two central bins for odd N). The constants of the large-N results below are not computed, so the following is open.

Conjecture 3.14 (convexity between the humps)**.** *For every $N > 2000$ the law at the crossing is convex between the humps.*

In double precision the law at the crossing is also convex between the humps for $N = 3200$, 6400, 12800 and 25600. The definition leaves out the two endpoints $a = 2m^*$ and $a = N - 2m^*$, because there the second difference is not positive for some $N \le 51$. Heuristically, between the humps $P_N(a)$ is close to $\frac{x}{2}\big[\frac{1}{a(a+1)} + \frac{1}{(N-a)(N-a+1)}\big]$, which is strictly convex in a. We also leave open the tail for m between about $x \log x$ and $x \log^2 x$, below the range of Theorem S2.1 of Online Resource 1.

Throughout this subsection $0 < \varepsilon < \frac{1}{2}$ and $c = 1 - 2\varepsilon$, so that $r_t(b) = \varepsilon + cb/t$. For $t \ge 1$ and every integer a we write $P_t(a) = \mathbb{P}(A_t = a)$, which is 0 unless $0 \le a \le t$, and

$$D_t(a) = P_t(a+1) - 2P_t(a) + P_t(a-1).$$

We also put $s_t = D_t(1) = P_t(0) - 2P_t(1) + P_t(2)$ for $t \ge 2$.

Lemma 3.15 (recurrence for the second difference)**.** *For every $t \ge 1$ and every integer a,*

$$P_{t+1}(a) = \big(1 - r_t(a)\big) P_t(a) + r_t(a-1)\, P_t(a-1), \tag{1}$$

$$D_{t+1}(a) = \big(1 - r_t(a+1)\big) D_t(a) + r_t(a-2)\, D_t(a-1). \tag{2}$$

Also $P_t(a) = P_t(t-a)$ *and* $D_t(a) = D_t(t-a)$. *The same two recurrences hold with* $f_t(m)$ *in place of* $P_t(a)$.

Proof. Given $A_t = a$, the copied step is $+1$ with probability a/t, so step $t+1$ is $+1$ with probability $(1-\varepsilon)\frac{a}{t} + \varepsilon(1-\frac{a}{t}) = r_t(a)$. This gives (1) for $0 \le a \le t+1$, where at $a = 0$ and at $a = t+1$ the formula has one extra term, namely $r_t(-1)P_t(-1)$ or $(1 - r_t(t+1))P_t(t+1)$, and this term is 0 because $P_t(-1) = P_t(t+1) = 0$. For other a every term vanishes. The symmetry of P_t is Lemma 2.2, and it gives the symmetry of D_t. The chain M of Lemma 2.2 has the same transition probabilities, so (1) holds for f_t as well.

For (2) put $\varsigma = c/t$, so that $r_t(b+1) = r_t(b) + \varsigma$. We insert (1) at $a+1$, a and $a-1$ into the definition of $D_{t+1}(a)$ and collect the coefficients of $P_t(a+1)$, $P_t(a)$, $P_t(a-1)$ and $P_t(a-2)$. They are

$$1 - r_t(a+1), \qquad 3r_t(a) - 2, \qquad 1 - 3r_t(a-1), \qquad r_t(a-2).$$

The right side of (2) has the coefficients $1-r_t(a+1)$, $-2+2r_t(a+1)+r_t(a-2)$, $1-r_t(a+1)-2r_t(a-2)$ and $r_t(a-2)$. The second ones agree because $2r_t(a+1) + r_t(a-2) = 3r_t(a)$, and the third ones agree because $r_t(a+1) + 2r_t(a-2) = 3r_t(a-1)$. □

Trinas and Valle [8, (2.2)] write (1) for the mass function of the position, which is the forward equation of the counting chain of Lemma 2.2, and they use it for a local limit theorem in the superdiffusive regime. In [8, (4.10)] they write the analogue of (2) for differences of higher order, and they use it at a fixed memory parameter in a certified study of log-concavity. We use (2) for signs. The coefficients in (2) are positive when $2 \le a \le t-1$. So the second difference moves like a mass that is partly killed, and its sign can change only through what enters at the two edges. The next lemma is the elementary form of the variation diminishing property that we need.

For a finite list z of real numbers let $\mathrm{sp}(z)$ be its sign pattern. We delete the zeros of z, replace each entry by its sign, and merge equal neighbors. So $\mathrm{sp}(z)$ is a list of alternating signs. For two such lists we write $p \preceq q$ if p is a subsequence of q. We put $\Pi_5 = (+,-,+,-,+)$ and $\Pi_3 = (+,-,+)$.

Lemma 3.16 (one step)**.** *Let* $\zeta = (\zeta_1, \dots, \zeta_n)$ *be real, and let* $\eta_i = A_i\zeta_i + B_i\zeta_{i+1}$ *for* $1 \le i \le n-1$ *with* $A_i \ge 0$ *and* $B_i \ge 0$. *Let* $w = (\zeta_1, \eta_1, \zeta_2, \eta_2, \dots, \eta_{n-1}, \zeta_n)$. *Then* $\mathrm{sp}(w) = \mathrm{sp}(\zeta)$ *and* $\mathrm{sp}(\eta) \preceq \mathrm{sp}(\zeta)$.

We prove this in Appendix D.1.

Lemma 3.17 (the sequence next to the end)**.** *For every* $t_1 \ge 2$ *we have* $\mathrm{sp}(s_2, s_3, \dots, s_{t_1}) \preceq \Pi_3$. *Also, for every* $0 \le \varepsilon \le 1$, *the numbers* s_3 *and* s_4 *do not both vanish.*

We prove this in Appendix D.2.

Theorem 3.18 (at most five blocks)**.** *Let* $N \ge 3$ *and* $0 < \varepsilon < \frac{1}{2}$. *Then*

$$\mathrm{sp}\big(D_N(1), D_N(2), \dots, D_N(N-1)\big) \preceq (+,-,+,-,+).$$

So the second difference of P_N *changes sign at most four times on* $\{1, \dots, N-1\}$. *The same holds for the second difference of* f_N, *and for* f_N *it holds for* $N = 2$ *as well.*

We prove this in Appendix D.3.

The next theorem is the step that turns one sign into a whole convex stretch. Its idea is short. A sign change from negative to positive at $a_0 < a_1$, together with its mirror image, gives the signs

$-,+,+,-$ at $a_0 < a_1 < N - a_1 < N - a_0$. A negative second difference in between would add a fifth alternating sign, which Theorem 3.18 forbids. A zero in between is subtler. We follow it back in time through (2), whose coefficients are positive, so a zero has two zero parents unless its parents have opposite signs, and opposite signs would again give five alternating signs. Going back to $t = 5$ forces $s_3 = s_4 = 0$, which Lemma 3.17 rules out.

Theorem 3.19 (no holes)**.** *Let $N \geq 5$ and $0 < \varepsilon < \frac{1}{2}$. Let $1 \leq a_0 < a_1 \leq N/2$ be integers with $D_N(a_0) < 0 < D_N(a_1)$. Then $D_N(a) > 0$ for every $a_1 \leq a \leq N - a_1$.*

We prove this in Appendix D.4.

Corollary 3.20 (one inequality is enough)**.** *Let $N \geq 8$ be even and $x = x_N$, or let $N \geq 5$ and $0 < \varepsilon < \frac{1}{2}$ be such that $P_N(0) < P_N(1)$. Let a^* be any mode of P_N and $m^* = N - a^*$. If $D_N(a_1) > 0$ for some integer a_1 with $m^* < a_1 \leq 2m^* + 1$ and $a_1 \leq N/2$, then $D_N(a) > 0$ for every $2m^* + 1 \leq a \leq N - 2m^* - 1$. In particular P_N is strictly convex on $[2m^*, N - 2m^*]$ as soon as $D_N(2m^* + 1) > 0$.*

Proof. At the crossing with $N \geq 8$ even, $P_N(0) = E_N < P_N(N-1) = P_N(1)$ by Proposition 3.7. Let a_0 be the smallest maximizer of P_N. Then $1 \leq a_0 \leq m^*$, because $P_N(0) < P_N(1)$ and $P_N(m^*) = P_N(a^*)$ is the maximum. Also $P_N(a_0 - 1) < P_N(a_0) \geq P_N(a_0 + 1)$, so $D_N(a_0) < 0$. Now Theorem 3.19 applies to $a_0 < a_1$. □

The condition $a_1 \leq N/2$ costs nothing at $a_1 = 2m^* + 1$, because $2m^* + 1 \leq N/2$ whenever the range $2m^* + 1 \leq a \leq N - 2m^* - 1$ is not empty. If $a^* < N/2$, then $m^* > N/2$, and no a_1 meets the conditions.

Theorem 3.21 (convexity between the humps for large N)**.** *There is N_1 such that the following holds for every $N \geq N_1$, even or odd, and every ε with $\log N \leq x \leq 2 \log N$. For every mode $a^* \geq N/2$ of P_N with $m^* = N - a^*$ we have $4m^* + 4 \leq N$ and*

$$P_N(a-1) - 2P_N(a) + P_N(a+1) > 0 \qquad \text{for } 2m^* + 1 \leq a \leq N - 2m^* - 1.$$

In particular, after enlarging N_1, the law at the crossing is convex between the humps for every even $N \geq N_1$. We do not compute the threshold N_1 (Remark 3.27).

Idea of the proof. By Corollary 3.20 it is enough to find one point of strict convexity between m^* and $2m^* + 1$. If there were none, P_N would be concave on $[m^*, 2m^* + 2]$, and then its value halfway would be at least half of its maximum. The Landau limit shows that the value halfway is much smaller.

Proof. Let N be large, and let $\log N \leq x \leq 2 \log N$. Then $\varepsilon \leq 2 \log N/N$, so (H) holds and $\varepsilon < \frac{1}{2}$. Also $(1 + \log x)/x \leq (1 + \log \log N)/\log N$, $x/N \leq 2 \log N/N$ and $x\delta_B = O(x^3 \log^2 N/N) = O(\log^5 N/N)$. So by Theorem 3.9(b) there are numbers $\Theta_N \to 0$, which depend on N alone, such that for every x in this range

$$\left| x f_N(m) - f_L(\lambda_m) \right| \leq \Theta_N \qquad \text{for every integer } m \geq 0. \tag{3}$$

The density f_L is continuous, because it is the Fourier transform of an integrable function. It is unimodal with mode λ_{L} (Lemma 3.10), so it is nonincreasing on $[\lambda_{\mathrm{L}}, \infty)$, and it is integrable. So $f_L(\lambda) \to 0$ as $\lambda \to \infty$. We write $o(1)$ for a quantity that is bounded in absolute value by a function of N alone that tends to 0. So every bound below holds uniformly in x.

Step 1. The mirror term. Let $0 \le m \le N/2$. Then $\lambda_{N-m} \ge \frac{N}{2x} - \log x \ge \frac{N}{4\log N} - \log(2\log N)$, which tends to ∞. So (3) gives $x f_N(N-m) \le f_L(\lambda_{N-m}) + \Theta_N = o(1)$. Since $P_N(m) = P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)]$, we get

$$2xP_N(m) = f_L(\lambda_m) + o(1) \qquad \text{uniformly for } 0 \le m \le N/2.$$

Step 2. The mode. Let m_0 be the integer part of $x\log x + \lambda_{\mathrm{L}} x$. Then $0 \le m_0 \le N/2$ and $|\lambda_{m_0} - \lambda_{\mathrm{L}}| \le 1/x$, so Step 1 and the continuity of f_L give $2xP_N(m_0) = f_L(\lambda_{\mathrm{L}}) + o(1)$. Now $P_N(m^*) \ge P_N(m_0)$ and $m^* \le N/2$, so Step 1 at m^* gives $f_L(\lambda_{m^*}) \ge f_L(\lambda_{\mathrm{L}}) - o(1)$. By Lemma 3.10(c) this forces $|\lambda_{m^*} - \lambda_{\mathrm{L}}| \le 1$ for large N. Hence

$$x\log x + (\lambda_{\mathrm{L}} - 1)x \le m^* \le x\log x + (\lambda_{\mathrm{L}} + 1)x, \qquad 2xP_N(m^*) \ge f_L(\lambda_{\mathrm{L}}) - o(1).$$

In particular $m^* \to \infty$ and $2m^* + 2 \le N/2$ for large N.

Step 3. Halfway. Let $v = m^* + \lfloor m^*/2 \rfloor$. Then $v \ge \frac{3}{2}m^* - 1$, so by Step 2

$$\lambda_v \ge \tfrac{1}{2}\log x + \tfrac{3}{2}(\lambda_{\mathrm{L}} - 1) - \tfrac{1}{x} \ge \tfrac{1}{2}\log\log N + \tfrac{3}{2}(\lambda_{\mathrm{L}} - 1) - 1,$$

which tends to ∞. Also $v \le N/2$. So Step 1 gives $2xP_N(v) = f_L(\lambda_v) + o(1) = o(1)$. Since $f_L(\lambda_{\mathrm{L}}) > 0$, Step 2 shows that $P_N(v) < \frac{1}{2}P_N(m^*)$ for large N.

Step 4. A point of convexity. Suppose that $D_N(a) \le 0$ for every $m^* < a \le 2m^* + 1$. Then P_N is concave on the integers of $[m^*, 2m^* + 2]$. With $u = m^*$ and $w = 2m^* + 2$ this gives

$$P_N(v) \ge \frac{w-v}{w-u}P_N(u) + \frac{v-u}{w-u}P_N(w) \ge \frac{w-v}{w-u}P_N(m^*) \ge \tfrac{1}{2}P_N(m^*),$$

because $w - v \ge \frac{1}{2}(m^* + 2) = \frac{1}{2}(w-u)$. This contradicts Step 3. So $D_N(a_1) > 0$ for some $m^* < a_1 \le 2m^* + 1$, and $a_1 \le N/2$ by Step 2. By Theorem 3.5, $P_N(1) = P_N(N-1) > \frac{x}{2}E_N > P_N(0)$, since $x \ge \log N > 2$. As $\varepsilon < \frac{1}{2}$, the second case of Corollary 3.20 applies and gives the claim. Finally, Theorem 2.7(b) gives $\log N \le x_N \le 2\log N$ for every large even N, which is the last claim. □

Lemma 3.22 (the center from convexity)**.** *Let $N \ge 8$ be even and $x = x_N$. Let $a^* \ge N/2$ be a mode of P_N and $m^* = N - a^*$. Suppose that $2m^* + 1 \le N/2$ and that $D_N(a) > 0$ for every $2m^* + 1 \le a \le N - 2m^* - 1$. Then $P_N(a) > C_N$ for every $1 \le a \le N - 1$ with $a \ne h$.*

Idea of the proof. Between the humps the law is strictly convex and symmetric, so the center is its only minimum there. Outside, a bin as low as the center would be a second valley. A valley needs a point of convexity with points of concavity on both sides, and with the mirror image this gives more changes of sign of the second difference than Theorem 3.18 allows.

Proof. Write $b = 2m^*$, so that $b + 1 \le h$, and put $\delta(a) = P_N(a+1) - P_N(a)$.

Between the humps. Since $D_N(a) = \delta(a) - \delta(a-1) > 0$ for $b + 1 \le a \le N - b - 1$, the differences $\delta(a)$ are strictly increasing for $b \le a \le N - b - 1$. By symmetry $\delta(h-1) = -\delta(h)$, and $\delta(h-1) < \delta(h)$ because $b \le h - 1$. So $\delta(h-1) < 0 < \delta(h)$. Hence P_N is strictly decreasing on $\{b, \dots, h\}$ and strictly increasing on $\{h, \dots, N-b\}$, and $P_N(a) > C_N$ for every $b \le a \le N - b$ with $a \ne h$. In particular $P_N(b) > C_N$.

Near the ends. By symmetry it is enough to treat $1 \le a \le b - 1$. Suppose that $P_N(a) \le C_N$ for some such a, and let k be the smallest index in $\{1, \dots, b\}$ at which P_N takes its least value on this set. By Proposition 3.7, $P_N(1) > P_N(0) = E_N = C_N$, and $P_N(b) > C_N$. So $2 \le k \le b - 1$,

and $P_N(k-1) > P_N(k) \le P_N(k+1)$, which gives $D_N(k) > 0$. Next let a' be the smallest index at which P_N takes its largest value on $\{0, \dots, k\}$. This value is at least $P_N(1)$, which exceeds both $P_N(0)$ and $P_N(k)$. So $1 \le a' \le k-1$ and $P_N(a'-1) < P_N(a') \ge P_N(a'+1)$, which gives $D_N(a') < 0$. In the same way let a'' be the largest index at which P_N takes its largest value on $\{k, \dots, h\}$. This value is at least $P_N(b)$, which exceeds both $P_N(k)$ and $P_N(h) = C_N$. So $k+1 \le a'' \le h-1$ and $P_N(a''-1) \le P_N(a'') > P_N(a''+1)$, which gives $D_N(a'') < 0$. Finally $D_N(h) > 0$ by hypothesis.

So D_N has the signs $-, +, -, +$ at $a' < k < a'' < h$. By the symmetry $D_N(a) = D_N(N-a)$ it has the signs $-, +, -$ at $N-a'' < N-k < N-a'$. Hence the sign pattern of $(D_N(1), \dots, D_N(N-1))$ has a subsequence of seven alternating signs. This contradicts Theorem 3.18, which allows at most five. □

Corollary 3.23 (the center is the unique interior minimum)**.** *For every large even N, at the crossing, $P_N(a) > C_N$ for every $1 \le a \le N-1$ with $a \ne h$. We do not compute the threshold for N.*

Proof. Let N be large and even, and let $a^* \ge N/2$ be a mode of P_N with $m^* = N - a^*$. By Corollary 3.11(b), $m^* = O(\log N \log\log N)$, so $2m^* + 1 \le N/2$ for large N. By Theorem 3.21, $D_N(a) > 0$ for $2m^* + 1 \le a \le N - 2m^* - 1$. So Lemma 3.22 applies. □

Proposition 3.24 (small N, computer-assisted)**.** *For every $5 \le N \le N_c = 2000$, even or odd, the law at the crossing is convex between the humps.*

Proof. The script `certify_convex.py` uses ball arithmetic with 160 bits (the library Arb). For each N it encloses ε_N in an interval of relative width at most $2 \cdot 10^{-8}$. It does so by evaluating the center and the end at the two ends of the interval, and Proposition 2.3 then places the crossing between them. It then runs (1) with ε equal to this interval. Each step only multiplies and adds enclosures of numbers in $[0, 1]$, so the radii stay small. From the resulting enclosure of P_N it lists every index that can be a mode, and for each of them it checks that the enclosure of $D_N(a)$ is positive for every $2m^* + 1 \le a \le N - 2m^* - 1$. When this range is not empty, so that $2m^* + 1 \le N/2$, it also checks that $D_N(a_0) < 0$ for some $1 \le a_0 \le 2m^*$ and that $D_N(2m^* + 1) > 0$. By Theorem 3.19 these two inequalities already imply the claim, so the test of every a is a second check. This covers $10 \le N \le 2000$.

For $5 \le N \le 9$ the script is not run, and the check is done in Lean (Section 4) by a computation in exact integer arithmetic with outward rounding, whose soundness Lean proves. Two rationals found by bisection in exact rational arithmetic bracket ε_N. Lean proves this by Proposition 2.3 from enclosures of $P_N(\lceil N/2 \rceil)$ and E_N at the two rationals. Then Lean encloses P_N for every ε between them at once. It checks that the upper rational is below $\frac{1}{2}$, that $P_N(0) < P_N(1)$, and that every index $a^* \ge N/2$ that can be a mode either has an empty range $2m^* + 1 \le a \le N - 2m^* - 1$ or has $2m^* + 1 \le N/2$ and $D_N(2m^* + 1) > 0$. By Corollary 3.20, with $a_1 = 2m^* + 1$, this gives the claim. At $N = 5$ the only index that can be a mode is $a^* = 4$, so the range is empty and the claim holds trivially. □

Proposition 3.25 (odd N)**.** *Let $N \ge 3$ be odd, and write $\widetilde{C}_N = P_N(\frac{N-1}{2}) = P_N(\frac{N+1}{2})$ for the two central bins. The ratio $\widetilde{C}_N / E_N$ is strictly increasing in ε, and it equals 1 at exactly one $\varepsilon_N \in (0, \frac{1}{2})$. Put $x_N = N\varepsilon_N$.*

(a) For every $0 < \varepsilon < \frac{1}{2}$,

$$\widetilde{C}_N = \Big(1 - \frac{c}{N}\Big)^{-1} C_{N+1},$$

where C_{N+1} is the center of the walk with $N+1$ steps and the same ε.

The center of a walk with an even number of steps is reached from the two central bins one step earlier, and from each of them with the same probability. So the two central bins of an odd N are the center of $N+1$ up to the factor $1-c/N$, and what we know about the crossing for even N carries over. Proposition S3.1 of Online Resource 1 shows that the crossing for odd N satisfies Theorem 2.7, that $P_N(0) < P_N(1)$ at the crossing for odd $N \geq 1689$, and that the law at the crossing is convex between the humps for every large odd N.

Proof. The crossing. By Proposition 2.3, $P_N(k)/E_N$ is strictly increasing in ε for $k = \frac{N+1}{2}$. It tends to 0 as $\varepsilon \to 0$. At $\varepsilon = \frac{1}{2}$ the steps are independent and fair, so $\widetilde{C}_N = \binom{N}{k} 2^{-N} > 2^{-N} = E_N$. So the ratio equals 1 at exactly one ε_N, and $\varepsilon_N < \frac{1}{2}$.

(a) Put $k = \frac{N+1}{2}$. Since $\varepsilon + \frac{c}{2} = \frac{1}{2}$, we have $r_N(k) = \frac{1}{2} + \frac{c}{2N}$ and $r_N(k-1) = \frac{1}{2} - \frac{c}{2N}$. So (1) with $t = N$ and $a = k$ gives

$$C_{N+1} = P_{N+1}(k) = \Big(\frac{1}{2} - \frac{c}{2N}\Big)\big(P_N(k) + P_N(k-1)\big) = \Big(1 - \frac{c}{N}\Big)\widetilde{C}_N.$$

□

Proposition 3.26 (the center for $8 \leq N \leq 2000$, computer-assisted)**.** *For every even $8 \leq N \leq 2000$, at the crossing, $P_N(a) > C_N$ for every $1 \leq a \leq N-1$ with $a \neq h$.*

Proof. Let $8 \leq N \leq 2000$ be even. By Proposition 3.24, $D_N(a) > 0$ for $2m^* + 1 \leq a \leq N - 2m^* - 1$ and every mode. For $N \geq 10$ the script `certify_center.py` encloses ε_N and the law P_N in the same way as `certify_convex.py`, lists every index $a^* \geq N/2$ that can be a mode, and checks that $2(N - a^*) + 1 \leq N/2$ for each of them. For $N = 8$ Lean checks the same, with an enclosure as in the proof of Proposition 3.24 for $N \leq 9$. This holds for every even N in the range. So Lemma 3.22 applies. □

As a second check, for each even $N \geq 10$ in the range the script also tests directly that the enclosure of $P_N(a) - P_N(h)$ is positive for every $1 \leq a \leq N-1$ with $a \neq h$, and this succeeds for all 996 values of N. The smallest value of $N/2 - (2m^* + 1)$ is 1, at $N = 8$ and at $N = 12$.

Remark 3.27 (what is still open). We do not compute the constant N_1 of Theorem 3.21. The constants of Theorem 3.9 are explicit, but the proof uses them only through a limit. Even if we followed them, Step 3 needs $f_L(\frac{3}{2}\lambda_{\mathrm{L}} + \frac{1}{2}\log x) < \frac{1}{2} f_L(\lambda_{\mathrm{L}})$ up to the error Θ_N. Numerically f_L falls to half of its maximum at $\lambda = 2.432$. So even without the error Θ_N, Step 3 needs $\frac{3}{2}\lambda_{\mathrm{L}} + \frac{1}{2}\log x > 2.432$, that is, $x > 252.8$. Since $x_N + \log x_N$ is close to $2\log N - \log 8$ (Theorem 2.7), this means that N is above about $3.5 \cdot 10^{56}$. So the N between N_{c} and N_1 (or the threshold N_1' of Proposition S3.1(c) of Online Resource 1 for odd N) are open, and no computation can reach these constants as the proof stands. By Corollary 3.20 (and Proposition S3.1(b) of Online Resource 1 for odd N) the whole range would be closed by an effective proof of the single inequality $D_N(a_1) > 0$ at one point $m^* < a_1 \leq 2m^* + 1$. Such a proof would need the second differences of P_N near $2m^*$ with an error below their size. Near $2m^*$ the Landau density suggests the size $3/[x^3(\log x)^4]$, since λ is close to $\log x$ there and $f_L''(\lambda) \sim 6/\lambda^4$ as $\lambda \to \infty$. This is a heuristic, not a local limit theorem for second differences. The main error in Theorem 3.9 is $(889 + 165\log x)/x$ for each value of $x f_N$, and this is far too large for the N in question.

4 Verification

The Python computations use two independent methods for the main checks (for example, forward and backward recursions for the elephant center agree to about 10^{-14} at $N = 12800$). Apart from short exact computations that can be checked by hand, such as the ratios C_N/E_N for $N = 2$, 4, 6 and 8 in Proposition 3.7, the proofs use computation in three places. First, some constants are values of explicit sums and integrals, such as the 60-term sum in the proof of Theorem 2.6(b), which we enclosed in ball arithmetic. It lies in $[102.586396, 102.586398]$, so the series is below the bound 102.59 used in the proof even with the tail added. Any finite bound suffices for the asymptotic statements. The value 5200 matters for the thresholds 1688 and 1689 in Proposition 3.7 and in Proposition S3.1(b) of Online Resource 1, and for the second-order bound there. Second, Lemma 3.10 rests on enclosures of f_L' at four points and of f_L'' on an interval, computed in ball arithmetic with rigorous error bounds. These two computations are in `certify_elephant.py`. Lean repeats the second one with enclosures of its own, as we describe below. Third, for every $10 \le N \le 2000$ the script `certify_convex.py` encloses the crossing ε_N in ball arithmetic and then encloses the whole law P_N at the crossing. Proposition 3.24 rests on this computation (with its output in `data/certify_convex_output_2026-10-17.txt`). The script `certify_center.py` repeats it for every even N in the range and proves Proposition 3.26, and Proposition 3.7 for $10 \le N < 1688$ (with its output in `data/certify_center_output_2026-10-17.txt`). For $5 \le N \le 9$, Propositions 3.24 and 3.26 rest instead on the checks in Lean described below. With the option `--witness` the two scripts also write a record for each N, with the rational ends of the enclosure of ε_N, the indices that can be a mode and the certified margins, under a header with the SHA-256 hashes of the scripts and the versions of Python and python-flint. The records of these runs are in `data/witnesses_convex_2026-10-17.jsonl` and `data/witnesses_center_2026-10-17.jsonl`. Each of the three `certify` scripts exits with nonzero status if any required certification fails, and the two scripts that take a range of N also refuse an empty range. The folder `data` also keeps the outputs of earlier runs, and the file `paperD/README.md` lists the statement numbers that they use. The tables, the figures, the numbers quoted outside the proofs and the evidence for Conjecture 3.14 beyond what Section 3.5 proves rest on computation alone. The script `verify_elephant_odd.py` checks the identity of Proposition 3.25(a) in exact rational arithmetic for six values of N, and it computes the numbers of Remark 3.27 for the Landau density in floating point.

Most statements of the paper and of Online Resource 1 are also proved in Lean 4.33.1 with Mathlib at commit `0df444a360ea`, in the folder `kagey-problems/problem131/lean/PaperD` (the file `MAP.md` there gives the Lean names and the status of each statement and the source of each classical input, and the file `lake-manifest.json` of the Lean project gives the full revision of Mathlib). The kernel checks every proof, and a recorded build of October 18, 2026, whose source hashes match the present files, checks every theorem and lemma of the library (which also holds the Lean files of a separate paper) and reports only the axioms `propext`, `Classical.choice` and `Quot.sound`.

Table 2 lists the results that Lean proves. Some of them rest on the six classical results of Table 3, which Lean states as named hypotheses, and nothing else is assumed. Where the notes give a wider range than the paper, we keep $0 < \varepsilon < 1$ because it defines the model, and $N \ge 16$ in Lemma S1.6 because Proposition S1.1 uses the lemma only for $N \ge 64$.

For Proposition 3.7 Lean checks $C_N < E_N$ at $\varepsilon = 2/(N+2)$ by an exact integer recursion for $8 \le N \le 200$, and it uses Theorem 2.6(b) for $N \ge 1688$. For $200 < N < 1688$, and for Propositions 3.24 and 3.26 over their whole range, Lean uses checkers whose soundness it proves. They enclose the law P_N in integer arithmetic with outward rounding, for every ε in a short

Table 2: The statements proved in Lean, with the abbreviations of Table 3.

Results	Status in Lean	Notes
Section 2		
Lemmas 2.2 and 2.4, Propositions 2.3 and 2.5, Theorem 1.1	proved	
Theorem 2.6	proved	By the argument of Appendix A, with (b) for every even $N \geq 2$ and the 60-term sum bounded by exact partial sums. The last display for $N \geq 64$ (for smaller N it follows from (a) and (b) by taking logarithms)
Theorem 2.7	proved	With the explicit bound for $L \geq 30$ in the proof
Section 3, finite shape and certificates		
Lemmas 3.1, 3.2 and 3.3	proved	For the random recursive tree with its marks, with the relabeling in Lemma 3.2
Lemmas 3.4, 3.15, 3.16, 3.17 and 3.22, Theorems 3.5, 3.18 and 3.19, Corollaries 3.6 and 3.20, Theorem 1.3(a), (b) and (d)	proved	
Propositions 3.7, 3.24 and 3.26	proved	By certificates, for every N in the range (see below)
Proposition 3.25, and Proposition S3.1(a), (b)	proved	With the explicit bound in S3.1(a). The last claim of S3.1(a) from LFI, CP, ES, UNI, as for Corollary 3.11(b)
Section 3, bulk and mode		
Theorem 3.8	proved from LFI, CP, ES	With the error δ_Σ, and also for $\varepsilon = 0$ and $\varepsilon = 1$
Theorem 3.9	local bounds from LFI, CP, ES, the limit in law also from SCH	With the constants 889, 165 and 6.6, and with f_L a probability density of a law of class L
Lemma 3.10	proved from UNI	Equation (8) by a rotation of the contour, and the integrals by an integration that the kernel checks, so Arb is not an input. Also $f_L(\lambda_{\mathrm{L}}) \geq 0.1806$, which we do not use
Corollary 3.11	(b) from LFI, CP, ES, UNI, and (a) in a different form	The form of (a) does the same job in the proofs
Theorem 3.21, Corollary 3.23, Proposition S3.1(c)	proved from LFI, CP, ES, UNI	For odd N Lean proves the conclusion of Corollary 3.23 only on $\{2m^*, \ldots, N - 2m^*\}$
Theorems 1.2 and 1.3(c)	proved from LFI, CP, ES, UNI, and SCH for the limit in law	
Appendices		
Lemmas A.1, A.2 and A.3	proved for the states that the argument uses	With Lemma A.1(a) read as a probability for every $c \geq 0$
Proposition A.4	proved as stated	Parts (c) and (d) for $0 < \varepsilon < \frac{1}{4}$ and every even $N \geq 2$, with the bound U_N
Appendices B and D	proved	With the constants 20, $\bar{c} = 1/40$ and $A_0 = 15400$, and Lemma B.2 and Proposition B.4 also for $0 \leq \varepsilon \leq 1$
Online Resource 1		
Proposition S1.4, Lemmas S1.5 and S1.6(a), (b)	proved	Proposition S1.4 for every real ε, and Lemma S1.6 under conditions weaker than $N \geq 16$
Propositions S1.1 and S1.2, Lemma S1.6(c), (d)	proved from HOEF	With the threshold $\log N \geq 3500$ of Proposition S1.2
Theorem S1.7	proved, with the Lean files of [10]	Those files prove its one input [10, Prop. 3.8], and a small library joins the two proofs
Section S2	proved	With the constants of Propositions S2.3 and S2.4. Theorem S2.1 needs $\varepsilon > 0$ and $x \geq 1$ only for large N
Section S3	proved	Part (c) of Proposition S3.1 as in the row above

Table 3: The classical results that Lean states as hypotheses.

Abbr.	Result	Source
LFI	Fourier inversion for an absolutely summable sequence on $\mathbb{Z}$	[30, Chapter I]
CP	characteristic function of $\sum_d d\mathcal{N}_d$, the $\mathcal{N}_d$ independent Poisson	[31, Section 3.3]
ES	inequality of Efron and Stein for independent coordinates	[27, 32]
SCH	lemma of Scheffé	[33]
UNI	unimodality of the laws of class L, used for f_L	[28]
HOEF	inequality of Hoeffding for the hypergeometric law	[34, Section 6]

interval at once. Two runs at the ends of the interval show that it contains ε_N, and a third run on the whole interval gives the inequalities. The ends come from the witness files (for $N \leq 9$ from a bisection in exact rational arithmetic). For $N > 100$ each row of the law is packed into one large integer, so that a step of the recursion takes a few operations on exact integers. The kernel evaluates every step of every run. For the longest runs the Lean files also give the rows at a few intermediate times, and the kernel recomputes each stretch between two of them and compares. A script chooses the precision of each run and writes these rows into the files, but nothing that it writes is trusted. So no proof rests on compiled code, and none uses `native_decide`.

Lean does not formalize the six classical results, the numerics (such as those of Remark 3.12) or Conjecture 3.14. The scripts in Python and Arb are separate from Lean. They ran under Python 3.14.3 with python-flint 0.9.0 (which contains FLINT 3.6.0 and with it Arb), mpmath 1.3.0, numpy 2.4.2 and scipy 1.17.1.

The scripts and the Lean files are in the repository named under "Code availability" in the Statements and Declarations, which also says how to rerun the checks.

Acknowledgments. I thank Peter Kagey for posing the problem and Aidan Deshong for insightful conversations.

A note on the use of AI

All mathematical claims, proofs and computations in this paper are my responsibility.

Statements and Declarations

Funding. No funding was received for this work.

Competing interests. The author has no competing interests to declare.

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

Data availability. The outputs of the computations described in Section 4 are in the folder `kagey-problems/problem131/paperD/data` of the repository named under Code availability. Online Resource 1 is the file `online_resource.pdf` in the folder `kagey-problems/problem131/paperD`, built from `online_resource.tex`.

Code availability. All code, data and Lean files supporting this paper are in the folder `kagey-problems/problem131` of https://github.com/apemm/Kagey-Problems. The supporting code, certificate records and Lean files are those at commit `4825316506a2` of that repository. To rerun the checks, run `python -B verify_all.py --paper d` in that folder, which takes several minutes. The certified computations behind Propositions 3.24 and 3.26 take about an hour and half an hour, and run separately in `paperD` as `python -B certify_convex.py 10 2000` and `python -B certify_center.py 10 2000`. The scripts need `numpy`, `scipy`, `mpmath` and `python-flint`, and the file `SUPPLEMENT.md` says what each script checks.

Use of AI tools. See the note on the use of AI tools just before these declarations.

Author contributions. I am the sole author and am responsible for the whole paper.

A Proofs for Section 2

We prove Theorem 2.6. Throughout, $N \geq 2$ is even and we condition on $X_1 = +1$. By Proposition 2.5(a) we need to bound $V_{t+1}(1)$, and we compare V with explicit hitting probabilities of Pólya urns. Let $\mathcal{L}_s F(b) = (1-r_s(b))F(b)+r_s(b)F(b+1)$, so that $V_s = \mathcal{L}_s V_{s+1}$ and $V_N(b) = \mathbf{1}\{b = h\}$. Both M_t and $t - M_t \geq 1$ are nondecreasing. So from $b \geq 1$ the chain stays at states ≥ 1, and for $1 \leq b \leq s-1$ we have $V_s(b) = 0$ unless b is feasible, that is, $b \in \mathcal{S}_s = \{b \in \mathbb{Z} : 1 \leq b \leq h,\ 1 \leq s - b \leq h\}$. Note that $\mathcal{S}_N = \{h\}$.

For $c \geq 0$ the Pólya(c) chain moves at time s from b to $b+1$ with probability $r_s^c(b) = \frac{b+c}{s+2c}$. It is Pólya's urn with weights $b+c$ and $s-b+c$ for the two colors, that is, with c extra balls of each color. For $b+c > 0$ and $s-b+c > 0$ let $\Phi_s^c(b) = \binom{N-s}{h-b}\mathrm{B}(h+c, h+c)/\mathrm{B}(b+c, s-b+c)$, where B is the beta function and the binomial coefficient is 0 unless $0 \leq h-b \leq N-s$. Theorem 2.6(a) will follow from Lemma A.1(d), and (b) from Proposition A.4.

Lemma A.1. *Let $c \geq 0$, $1 \leq s \leq N$, $b+c > 0$ and $s-b+c > 0$.*

(a) The number $\Phi_s^c(b)$ is the probability that the Pólya(c) chain at b at time s is at h at time N. So $\Phi_N^c(b) = \mathbf{1}\{b = h\}$, and $\Phi_s^c(b) = r_s^c(b)\Phi_{s+1}^c(b+1) + (1 - r_s^c(b))\Phi_{s+1}^c(b)$ for $s < N$.

(b) If $s < N$ and $\Phi_s^c(b) > 0$, then $r_s^c(b)\Phi_{s+1}^c(b+1)/\Phi_s^c(b) = \bar{r}_s(b) := \frac{h-b}{N-s}$.

(c) If $0 < \varepsilon < \frac{1}{2}$ and $\alpha = \frac{\varepsilon}{1-2\varepsilon}$, then $r_s(b) = r_s^{\alpha s}(b)$ for $0 \leq b \leq s$.

(d) If $0 < \varepsilon < \frac{1}{2}$ and $0 \leq c \leq \alpha s$, then $V_s(b) \geq \Phi_s^c(b)$ for $1 \leq b \leq s-1$. In particular $V_{t+1}(1) \geq \Phi_{t+1}^0(1) = \pi_t$ for $1 \leq t \leq h$.

Proof. (a) Let $x = b+c$, $y = s-b+c$ and $(x)_k = x(x+1)\cdots(x+k-1)$. As in the proof of Lemma 2.4, each order of the $N-s$ draws with $h-b$ draws of the first color has probability $(x)_{h-b}(y)_{N-s-h+b}/(x+y)_{N-s} = \mathrm{B}(h+c, h+c)/\mathrm{B}(x, y)$, and there are $\binom{N-s}{h-b}$ such orders. The recursion is the Markov property.

(b) Use $\binom{N-s-1}{h-b-1}/\binom{N-s}{h-b} = \frac{h-b}{N-s}$ and $\mathrm{B}(x, y)/\mathrm{B}(x+1, y) = \frac{x+y}{x} = 1/r_s^c(b)$.

(c) Since $1 + 2\alpha = 1/(1-2\varepsilon)$, we have $\frac{b+\alpha s}{s+2\alpha s} = (1-2\varepsilon)(\frac{b}{s} + \alpha) = r_s(b)$.

(d) We check the hypotheses of Lemma A.2(b) below for $F_u = \Phi_u^c$ on $\{1, \ldots, u-1\}$, where $s \leq u \leq N$. The reason is that M has $\alpha u \geq c$ extra balls at time u, so it is pulled toward $\frac{u}{2}$ at least as strongly as the Pólya(c) chain, and from a state closer to $\frac{u}{2}$ it is easier to end at h.

By (a), $F_N(b) = \mathbf{1}\{b = h\}$, and F_u vanishes off $\mathcal{S}_u$ because the binomial coefficient does. Let $s \le u < N$ and $b \in \mathcal{S}_u$. Subtracting the recursion in (a) from the definition of $\mathcal{L}_u$ gives

$$\mathcal{L}_u F_{u+1}(b) - F_u(b) = \big(r_u(b) - r_u^c(b)\big)\big(F_{u+1}(b+1) - F_{u+1}(b)\big).$$

We show that the two factors on the right never have opposite signs. For the first factor, $r_u^c(b) = \frac{1}{2} - (\frac{u}{2} - b)/(u + 2c)$, and by (c) the same formula with αu in place of c gives $r_u(b)$. So the first factor is $\frac{u}{2} - b$ times $\frac{1}{u+2c} - \frac{1}{u+2\alpha u}$. The second of these is nonnegative because $c \le \alpha s \le \alpha u$. Hence the first factor has the sign of $\frac{u}{2} - b$ or vanishes. For the second factor there are three cases. If $b = h$ then $b \ge \frac{u}{2}$ and $F_{u+1}(b+1) = 0$, so the factor is ≤ 0. If $u - b = h$ then $b \le \frac{u}{2}$ and $F_{u+1}(b) = 0$, so the factor is ≥ 0. Otherwise both values are positive, and the formula for Φ gives $F_{u+1}(b+1)/F_{u+1}(b) = \frac{(h-b)(u-b+c)}{(h-u+b)(b+c)}$. With $b = \frac{u}{2} - d$ the numerator minus the denominator is $2d(h + c)$, so the ratio is ≥ 1 if $b \le \frac{u}{2}$ and ≤ 1 if $b \ge \frac{u}{2}$. In all three cases the second factor has the sign of $\frac{u}{2} - b$ or vanishes. Hence $\mathcal{L}_u F_{u+1} \ge F_u$ on $\mathcal{S}_u$, and Lemma A.2(b) with $s_0 = s$ gives $V_s \ge F_s$. Finally $\Phi_{t+1}^0(1) = \beta_{t+1}(h) = \pi_t$ by (a) and Lemma 2.4. □

Lemma A.2. *Let $1 \le s_0 \le N$, and for $s_0 \le s \le N$ let $F_s \ge 0$ be a function on $\{1, \dots, s-1\}$.*

(a) If $F_N(h) \ge 1$ and $\mathcal{L}_s F_{s+1} \le K_s F_s$ on $\mathcal{S}_s$ for $s_0 \le s < N$, with constants $K_s \ge 0$, then $V_s \le \big(\prod_{u=s}^{N-1} K_u\big) F_s$ on $\mathcal{S}_s$.

(b) If $F_N(b) \le \mathbf{1}\{b = h\}$, each F_s vanishes off $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} \ge F_s$ on $\mathcal{S}_s$ for $s_0 \le s < N$, then $V_s \ge F_s$.

Proof. We use backward induction on s, from $V_N(b) = \mathbf{1}\{b = h\}$ and $V_s = \mathcal{L}_s V_{s+1}$. Note that $\mathcal{L}_s$ is an average of two values, so it keeps inequalities between functions. In (a), suppose that $V_{s+1} \le \prod_{u>s} K_u\, F_{s+1}$ on $\mathcal{S}_{s+1}$. This also holds off $\mathcal{S}_{s+1}$, because there $V_{s+1} = 0$ and $F_{s+1} \ge 0$. So on $\mathcal{S}_s$ we get $V_s = \mathcal{L}_s V_{s+1} \le \prod_{u>s} K_u\, \mathcal{L}_s F_{s+1} \le \prod_{u \ge s} K_u\, F_s$. In (b), suppose that $V_{s+1} \ge F_{s+1}$ on $\{1, \dots, s\}$. Then $V_s = \mathcal{L}_s V_{s+1} \ge \mathcal{L}_s F_{s+1} \ge F_s$ on $\mathcal{S}_s$, and $F_s = 0 \le V_s$ off $\mathcal{S}_s$. □

Idea of the proof of the upper bound. We compare M with an urn whose number of extra balls, $\kappa(s + 10)$ of each color at time s, grows more than twice as fast as the αs of M. Its hitting probability $\Phi_s^{\kappa_s}$ is explicit. One step of M applied to it gives it back times a factor Δ_s, and Proposition A.4 shows that $\log \Delta_s \le \kappa/s$ up to a smaller term. Two effects enter Δ_s. Raising the extra balls from κ_s to κ_{s+1} changes Φ, and we bound this through the derivative of Φ_s^τ in τ (Lemma A.3). Also M pulls toward the middle less than the comparison urn, which helps. The product of the factors is about $e^{\kappa H_N}$. Let ψ be the digamma function.

Lemma A.3. *Let $2 \le s \le N$, $b \in \mathcal{S}_s$ and $\tau \ge 0$. Put $X = b + \tau$, $Y = s - b + \tau$, $\Sigma = X + Y$ and $z = (s - 2b)^2/\Sigma^2$. Then*

$$\partial_\tau \log \Phi_s^\tau(b) < -\log(1 - z) + \frac{1 + z}{(1 - z)\Sigma} + \frac{2(1 + z)}{3\Sigma^2(1 - z)^2}.$$

Proof. We differentiate the beta functions and bound the digamma function by the first terms of its expansion. Only the two beta functions in $\Phi_s^\tau(b)$ depend on τ, and $\log \mathrm{B}(x, y) = \log \Gamma(x) + \log \Gamma(y) - \log \Gamma(x + y)$. So the derivative is $2\psi(\Sigma) - \psi(X) - \psi(Y)$ minus the difference $2\psi(N + 2\tau) - 2\psi(h + \tau)$. The duplication formula [35, 5.5.5] gives $2\psi(2y) = \psi(y) + \psi(y + \frac{1}{2}) + 2\log 2$, and ψ is increasing. With $y = h + \tau$ this shows that the difference we subtract is more than $2 \log 2$. For $x > 0$, [35, 5.9.15] writes $\log x - \frac{1}{2x} - \psi(x)$ as $2\int_0^\infty \frac{t\,dt}{(t^2 + x^2)(e^{2\pi t} - 1)}$, which lies in $(0, \frac{1}{12x^2})$ since $\int_0^\infty \frac{t\,dt}{e^{2\pi t} - 1} = \frac{1}{24}$.

So $\psi(\Sigma) < \log\Sigma - \frac{1}{2\Sigma}$, and $\psi(x) > \log x - \frac{1}{2x} - \frac{1}{12x^2}$ for $x = X$ and $x = Y$. Hence the derivative is less than $\log\frac{\Sigma^2}{4XY} + (\frac{1}{2X} + \frac{1}{2Y} - \frac{1}{\Sigma}) + \frac{1}{12}(\frac{1}{X^2} + \frac{1}{Y^2})$. Now $X - Y = 2b - s$, so $4XY = \Sigma^2 - (X-Y)^2 = \Sigma^2(1-z)$ and $X^2 + Y^2 = \Sigma^2(1+z)/2$. So the three terms are $-\log(1-z)$, $\frac{X^2+Y^2}{2XY\Sigma} = \frac{1+z}{(1-z)\Sigma}$ and $\frac{X^2+Y^2}{12X^2Y^2} = \frac{2(1+z)}{3\Sigma^2(1-z)^2}$. □

Proposition A.4. *Let $0 < \varepsilon < \frac{1}{4}$, $\kappa = \frac{2\varepsilon}{1-4\varepsilon}$, $\kappa_s = \kappa(s+10)$ and $\Sigma_s = s + 2\kappa_s$. For $1 \le s \le N-1$ and $b \in \mathcal{S}_s$ let $\Delta_s(b) = \mathcal{L}_s\Phi_{s+1}^{\kappa_{s+1}}(b)/\Phi_s^{\kappa_s}(b)$.*

(a) Let $c = \kappa_s$, $c' = \kappa_{s+1}$, $d = \frac{s}{2} - b$, $u = \frac{d}{s+2c'}$, $v = \frac{d}{N-s}$ and $\rho = \frac{s+2c'}{s(1+2\alpha)}$. Then $|u| < \frac{1}{2}$ and

$$\Delta_s(b) = \mathcal{J}\,\frac{\Phi_s^{c'}(b)}{\Phi_s^{c}(b)}, \qquad \mathcal{J} = 1 - (\rho-1)\frac{u(u+v)}{\frac{1}{4} - u^2}.$$

(b) The factor satisfies $\log\Delta_s(b) \le \frac{\kappa}{\Sigma_s} + \frac{2\kappa}{3\Sigma_s^2}$.

(c) For $1 \le t \le h$, $V_{t+1}(1) \le \pi_t \exp\big(\kappa\big[H_{N-1} - H_t + \frac{2}{3t}\big] + \kappa_{t+1}\big[\log\frac{(t+1)^2}{4t} + \frac{2}{3}\big]\big)$.

(d) The center satisfies $C_N \le U_N := \varepsilon c_1 e^{\kappa H_{N-1}} \sum_{t=1}^{h} w_t(1-\varepsilon)^{t-1} e^{\kappa\mathcal{G}(t)}$, where $\mathcal{G}(t) = (t + 11)\big(\log\frac{(t+1)^2}{4t} + \frac{2}{3}\big) - H_t + \frac{2}{3t}$.

The slope κ solves $2(\kappa-\alpha)/(1+2\alpha) = \kappa$, so that in (b) the gain in $\mathcal{J}$ cancels the main loss from raising the extra balls from c to c'.

Proof. (a) Write $r' = r_s^{c'}(b)$, $\bar r = \bar r_s(b)$ and $r = r_s(b)$. By Lemma A.1(b) with c', and the recursion in Lemma A.1(a), we have $r'\Phi_{s+1}^{c'}(b+1) = \bar r\,\Phi_s^{c'}(b)$ and $(1-r')\Phi_{s+1}^{c'}(b) = (1-\bar r)\Phi_s^{c'}(b)$. So the numerator of $\Delta_s(b)$ divided by $\Phi_s^{c'}(b)$ is $\frac{(1-r)(1-\bar r)}{1-r'} + \frac{r\bar r}{r'}$. Here $r' = \frac{1}{2} - u$, $\bar r = \frac{1}{2} + v$ and $r = \frac{1}{2} - \rho u$, since $\frac{1}{2} - r = (1-2\varepsilon)\frac{d}{s} = \rho u$. Over the denominator $r'(1-r') = \frac{1}{4} - u^2$ the numerator is $f(u,v) + f(-u,-v)$ with $f(u,v) = (\frac{1}{2} + \rho u)(\frac{1}{2} - v)(\frac{1}{2} - u)$, and this equals $\frac{1}{4} - \rho u^2 - (\rho-1)uv$. Also $|d| \le \frac{s}{2} - 1$, so $|u| < \frac{1}{2}$.

(b) *Step 1. The factor $\mathcal{J}$.* Write $c = \kappa_s$, $c' = c + \kappa$ and $\Sigma = \Sigma_s$. Since $c' - \alpha s = (\kappa - \alpha)s + 11\kappa$, the choice of κ gives $\rho - 1 = \kappa + \nu_s$ with $\nu_s = \frac{22\kappa}{s(1+2\alpha)}$. As u and v have the sign of d, $u(u+v) \ge u^2$. So $\mathcal{J} \le 1 - (\kappa+\nu_s)W'$ with $z' = 4u^2$ and $W' = \frac{z'}{1-z'}$. Also $\mathcal{J} > 0$ by (a), since $0 < r < 1$ and $\bar r$, $1 - \bar r$ are not both 0. So $\log\mathcal{J} \le -(\kappa+\nu_s)W'$.

Step 2. The ratio $\Phi_s^{c'}/\Phi_s^{c}$. The logarithm $\log(\Phi_s^{c'}(b)/\Phi_s^{c}(b))$ is the integral over $\tau \in [c, c']$ of the derivative in Lemma A.3. Its bound increases in z and decreases in Σ, while z decreases and Σ increases in τ. So we may take the bound at $\tau = c$, where $X + Y = \Sigma$, and multiply it by the length κ of the interval. Put $W = \frac{z}{1-z}$. Then $\frac{1}{1-z} = 1 + W$, $\frac{1+z}{1-z} = 1 + 2W$ and $-\log(1-z) \le W$, so

$$\log\Delta_s(b) \le \kappa\Big[W + \frac{1+2W}{\Sigma} + \frac{2(1+2W)(1+W)}{3\Sigma^2}\Big] - (\kappa+\nu_s)W'. \tag{4}$$

Step 3. What is left to show. The terms of (4) without W and W' are $\frac{\kappa}{\Sigma} + \frac{2\kappa}{3\Sigma^2}$, which is the bound we want. Since $(1+2W)(1+W) = 1 + 3W + 2W^2$, the other terms are

$$\kappa(W - W') + \frac{2\kappa W}{\Sigma} + \frac{2\kappa W}{\Sigma^2} + \frac{4\kappa W^2}{3\Sigma^2} - \nu_s W',$$

and we must show that they add up to at most 0. If $W = 0$ then $d = 0$, so $W' = 0$ and there is nothing to do. So let $W > 0$. The first four terms are positive, and we show that each is a small

multiple of $\frac{\kappa}{s}W$. The last term is negative (it comes from the shift 10 in κ_s), and it is a larger multiple of $\frac{\kappa}{s}W$. We need three facts, with $X = b + c$ and $Y = s - b + c$ as in Step 2.

(i) We have $1 + W \le \frac{\Sigma}{2(1+c)}$. Indeed $X, Y \ge 1 + c$, since $b \ge 1$ and $s - b \ge 1$, and the larger of X and Y is at least $\frac{\Sigma}{2}$. So $XY \ge (1 + c)\Sigma/2$, and $1 + W = \frac{1}{1-z} = \frac{\Sigma^2}{4XY}$.

(ii) We have $W - W' \le \frac{2\kappa}{1+c}W$, and so $W' \ge \frac{9}{11}W$. Indeed z and z' have the same numerator $(s-2b)^2$, and their denominators are Σ^2 and $(s+2c')^2 = (\Sigma+2\kappa)^2$. So $z' = \lambda z$ with $\lambda = (\frac{\Sigma}{\Sigma+2\kappa})^2 \ge 1 - \frac{4\kappa}{\Sigma}$. Then

$$W - W' = \frac{(1-\lambda)z}{(1-z)(1-\lambda z)} \le (1-\lambda)W(1+W) \le \frac{4\kappa}{\Sigma}\, W\, \frac{\Sigma}{2(1+c)} = \frac{2\kappa}{1+c}W,$$

where we used $1 - \lambda z \ge 1 - z$ and then (i). Since $c = \kappa(s + 10) \ge 11\kappa$, we have $\frac{2\kappa}{1+c} \le \frac{2}{11}$.

(iii) We have $\Sigma \ge s$, $1 + c > \kappa s$ and $\Sigma^2 \ge 3s$. The first two are clear. For the third, $W > 0$ means $s \ne 2b$, and with $b \ge 1$ and $s - b \ge 1$ this forces $s \ge 3$.

Step 4. The terms with W. We bound the four positive terms one by one.

$$\begin{aligned}
\kappa(W - W') &\le \frac{2\kappa^2}{1+c}W \le \frac{2\kappa}{s}W && \text{by (ii) and } 1 + c > \kappa s,\\
\frac{2\kappa W}{\Sigma} &\le \frac{2\kappa}{s}W && \text{by } \Sigma \ge s,\\
\frac{2\kappa W}{\Sigma^2} &\le \frac{2\kappa}{3s}W && \text{by } \Sigma^2 \ge 3s,\\
\frac{4\kappa W^2}{3\Sigma^2} &\le \frac{4\kappa W(1+W)}{3\Sigma^2} \le \frac{2\kappa W}{3\Sigma(1+c)} \le \frac{2\kappa}{3s}W && \text{by (i) and } \Sigma \ge s.
\end{aligned}$$

The four bounds add up to $\frac{16\kappa}{3s}W$. For the negative term, (ii) gives $-\nu_s W' \le -\frac{9}{11}\nu_s W = -\frac{18\kappa}{s(1+2\alpha)}W$. So

$$\log \Delta_s(b) \le \frac{\kappa}{\Sigma} + \frac{2\kappa}{3\Sigma^2} + W\Big[\frac{16\kappa}{3s} - \frac{18\kappa}{s(1+2\alpha)}\Big],$$

and the bracket is negative because $1 + 2\alpha < 2$ for $\varepsilon < \frac{1}{4}$, so $18/(1 + 2\alpha) > 9 > 16/3$.

(c) We first compare $V_{t+1}(1)$ with $\Phi_{t+1}^{\kappa_{t+1}}(1)$, and then this with $\pi_t = \Phi_{t+1}^{0}(1)$. For the first step apply Lemma A.2(a) with $s_0 = t + 1$, $F_s = \Phi_s^{\kappa_s}$ and $K_s = \exp(\frac{\kappa}{s} + \frac{2\kappa}{3s^2})$. This is allowed, since $F_N(h) = 1$, $F_s > 0$ on $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} = \Delta_s F_s \le K_s F_s$ on $\mathcal{S}_s$ by (b) and $\Sigma_s \ge s$. The product of the K_s over $t < s < N$ is the exponential of $\kappa(H_{N-1} - H_t) + \frac{2\kappa}{3}\sum_{s>t} s^{-2}$, and $\sum_{s>t} s^{-2} \le \frac{1}{t}$. Since $1 \in \mathcal{S}_{t+1}$, we get $V_{t+1}(1) \le e^{\kappa[H_{N-1} - H_t + 2/(3t)]}\, \Phi_{t+1}^{\kappa_{t+1}}(1)$.

For the second step, $\log(\Phi_{t+1}^{\kappa_{t+1}}(1)/\pi_t)$ is the integral over $\tau \in [0, \kappa_{t+1}]$ of the derivative in Lemma A.3 with $s = t+1$ and $b = 1$. As in Step 2 of (b), the bound of that lemma is largest at $\tau = 0$. There $X = 1$, $Y = t$ and $\frac{1}{1-z} = \frac{(t+1)^2}{4t}$, and the bound is $\log \frac{(t+1)^2}{4t} + \frac{t^2+1}{2t(t+1)} + \frac{t^2+1}{12t^2} \le \log \frac{(t+1)^2}{4t} + \frac{2}{3}$, because the two fractions are at most $\frac{1}{2}$ and $\frac{1}{6}$ for $t \ge 1$. Multiplying by the length κ_{t+1} of the interval gives (c).

(d) Insert (c) into Proposition 2.5(a), and use $\kappa_{t+1} = \kappa(t + 11)$ and $\pi_t = c_1 w_t$. □

Proof of Theorem 2.6. (a) By Proposition 2.5(a) and Lemma A.1(d), $C_N \ge \varepsilon c_1 \sum_t w_t(1 - \varepsilon)^{t-1}$. Now $(1-\varepsilon)^{t-1} \ge 1-(t-1)\varepsilon$ and $\sum_t w_t(t-1) = 2 - \frac{6}{h+2}$ by Proposition 2.5(c). So $C_N \ge \varepsilon c_1(1-2\varepsilon)$, and $c_1 = \frac{4}{N+2}$.

(b) *Step 1. The constants.* Let $0 < \varepsilon(1 + \log N) \le \frac{1}{100}$. Since $\log N \ge 0$, we have $\varepsilon \le \frac{1}{100}$. So $1-4\varepsilon \ge 0.96$ and $\kappa \le 2\varepsilon/0.96 \le 2.1\varepsilon$. Also $\kappa \le 2\varepsilon(1+8\varepsilon)$, because $(1+8\varepsilon)(1-4\varepsilon) = 1+4\varepsilon(1-8\varepsilon) \ge$

1. Since $\log\frac{(t+1)^2}{4t} \ge 0$ and $\frac{2}{3}(t+11) > 1+\log t \ge H_t$, we have $\mathcal{G}(t) > 0$. Since $\frac{(t+1)^2}{4t} \le \frac{t+1}{2}$, $\frac{2}{3} < \log 2$ and $H_t \ge \frac{2}{3t}$, also $\mathcal{G}(t) \le (t+11)\log(t+1) \le 12t\log(t+1)$. So $\kappa\mathcal{G}(t) \le 25.2\,\varepsilon t\log(t+1)$, and this is at most $\frac{t}{2}\log 2$ for $t \le h$, since then $25.2\,\varepsilon\log(t+1) \le 25.2\,\varepsilon\log N \le 0.252 < \frac{1}{2}\log 2$.

Step 2. The sum over t. Since $e^y - 1 \le ye^y$ for $y \ge 0$, Step 1 gives $e^{\kappa\mathcal{G}(t)} - 1 \le \kappa\mathcal{G}(t)2^{t/2}$. By Proposition A.4(d), $(1-\varepsilon)^{t-1} \le 1$, $\sum_t w_t = 1$ and $w_t \le 2t2^{-t}$ (Proposition 2.5(c)),

$$C_N \le \varepsilon c_1 e^{\kappa H_{N-1}}\Big(1 + 50.4\,\varepsilon\sum_{t\ge1} t^2\log(t+1)2^{-t/2}\Big),$$

and we bound the series as follows. The ratio of its consecutive terms is $(1+\frac{1}{t})^2\frac{\log(t+2)}{\log(t+1)}2^{-1/2}$, which decreases in t and is below $\frac{3}{4}$ at $t = 60$. So the terms with $t > 60$ add up to less than four times the term at $t = 61$, and this is below 10^{-4}. The first 60 terms add up to 102.5864 to four decimals (enclosed in ball arithmetic by `certify_elephant.py`), so the series is below 102.59.

Step 3. The factor $e^{\kappa H_{N-1}}$. Finally $H_{N-1} \le 1+\log N$, so $\kappa H_{N-1} \le 2\varepsilon\log N + 2\varepsilon + 16\varepsilon^2(1+\log N) \le 2\varepsilon\log N + 3\varepsilon$. With $1+y \le e^y$ this proves (b), since $3 + 50.4\cdot 102.59 < 5200$. Any upper bound for the series below $(5200-3)/50.4$, which exceeds 103.1, would give the same constant. □

B The families and the mode

This appendix proves the bound on the mode in Lemma B.5, which Corollary 3.11 uses. Online Resource 1 uses the same lemmas for the tail. We keep the conventions of Section 3. All constants in this appendix are explicit. Online Resource 1 writes C for an absolute constant whose value may change from line to line. Since M is independent of X_1, $f_N(m) = \mathbb{P}(M_N = m)$. Every path of a chain $M^{[k_0]}$ is nondecreasing in time, and we use the paths of Y and $Z^{(j)}$ fixed after Lemma 3.3. The size $|\mathcal{T}_v|$ of the family has the law β_v of Lemma 2.4. Let $\bar\beta_v(s) = \mathbb{P}(|\mathcal{T}_v| \ge s)$, with $\bar\beta_v(s) = 1$ for $s \le 1$. By Pascal's rule $\binom{N-s}{v-1} - \binom{N-s-1}{v-1} = \binom{N-s-1}{v-2}$ the tail of β_v telescopes. So, for $1 \le s \le N-1$,

$$\bar\beta_v(s) = \binom{N-s}{v-1}\Big/\binom{N-1}{v-1} = \prod_{i=1}^{v-1}\Big(1-\frac{s-1}{N-i}\Big) \le e^{-(v-1)(s-1)/(N-1)}, \qquad \beta_v(s) = \frac{v-1}{N-s}\,\bar\beta_v(s), \tag{5}$$

where the product vanishes as soon as one factor does. Where they are defined, the ratios are

$$\frac{\beta_v(d)}{\beta_v(d-1)} = 1 - \frac{v-2}{N-d}, \qquad \frac{\bar\beta_v(s-1)}{\bar\beta_v(s)} = 1 + \frac{v-1}{N-s-v+2}. \tag{6}$$

Lemma B.1 (monotone sandwich)**.** *Let $0 \le a \le b \le N$. If $m_h \le a$, then $(b-a+1)f_N(b) \le \mathbb{P}(a \le M_N \le b)$. If $b \le m_h$, then $(b-a+1)f_N(a) \le \mathbb{P}(a \le M_N \le b)$.*

Proof. By Lemma 3.4, $f_N(\ell) \ge f_N(b)$ for $a \le \ell \le b$ in the first case, and $f_N(\ell) \ge f_N(a)$ in the second. Sum over ℓ. □

Lemma B.2 (tail)**.** *Let $0 < \varepsilon < 1$, $k_0 \ge 1$ and $t \ge k_0$. For every integer $z \ge 1$,*

$$\mathbb{P}\big(M_t^{[k_0]} \ge z\big) \le \frac{\varepsilon t H_z}{z}, \qquad \mathbb{E}M_t^{[k_0]} \le \varepsilon t\log\frac{t}{k_0}.$$

In particular $\mathbb{P}(M_N \ge z) \le xH_z/z$, and $\mathbb{P}(Z_s^{(j)} \ge z) \le xH_z/z$ for $j-1 \le s \le N$.

Proof. We use the tree of Lemma 3.1 on $[t]$, with marks only on $v > k_0$. Every odd vertex lies in the family of a marked vertex, so $M_t^{[k_0]} \le \sum_{v \text{ marked}} |\mathcal{T}_v|$. We call a family large if its size is at least z, and small otherwise. If $M_t^{[k_0]} \ge z$, then some marked family is large, or the small marked families have total size at least z. We use the union bound for the first event and Markov's inequality for the second. Marks and tree are independent, so

$$\mathbb{P}\big(M_t^{[k_0]} \ge z\big) \le \varepsilon \sum_v \mathbb{P}(|\mathcal{T}_v| \ge z) + \frac{\varepsilon}{z} \sum_v \mathbb{E}\big[|\mathcal{T}_v|; |\mathcal{T}_v| < z\big].$$

By the first sum in Lemma 2.4, with t in place of N, on average $\frac{t}{d(d+1)}$ vertices have a family of size d. So the first sum over v is $\sum_{d \ge z} \frac{t}{d(d+1)} \le \frac{t}{z}$, and the second is $\sum_{d=1}^{z-1} \frac{t}{d+1} = t(H_z - 1)$. Together they give $\frac{\varepsilon t H_z}{z}$. For the mean, only the vertices $v > k_0$ can be marked and $\mathbb{E}|\mathcal{T}_v| = t/v$, so $\mathbb{E}M_t^{[k_0]} \le \varepsilon \sum_{v=k_0+1}^{t} \frac{t}{v} \le \varepsilon t \log \frac{t}{k_0}$. □

Lemma B.3 (the first flipped family)**.** *For $1 \le d \le N-1$,*

$$\frac{x}{d(d+1)}\Big(1 - \frac{2x(N-d-1)}{N(d+2)}\Big) \le \mathbb{P}\big(|\mathcal{T}_V| = d\big) \le \frac{x}{d(d+1)}, \qquad \mathbb{P}\big(|\mathcal{T}_V| \ge d\big) \le x\Big(\frac{1}{d} - \frac{1}{N}\Big).$$

Proof. By Lemma 3.3, $\mathbb{P}(|\mathcal{T}_V| = d) = \sum_j \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and $1 - (j-2)\varepsilon \le (1-\varepsilon)^{j-2} \le 1$. Apply the two sums of Lemma 2.4, and sum the upper bound over $d' \ge d$. □

Proposition B.4 (local lower bound)**.** *Let $N \ge 64$, $0 < \varepsilon < 1$ and $2 \le m \le N/4$. Then $f_N(m) \ge \frac{x}{m(m+1)}(1 - e_1'(m)) \ge \frac{x}{m(m+1)}(1 - e_1(m))$, where*

$$e_1'(m) = 2\varepsilon H_{m+1} + \frac{xH_m}{m} + \frac{2x}{m+2} + 8\varepsilon \log N + 20me^{-m/9}$$

and

$$e_1(m) = 20\Big(\frac{x \log(m+1)}{m} + \varepsilon \log N + me^{-m/9}\Big).$$

Idea of the proof. We use the formula of Lemma 3.3. Given the time j of the first flip, the count M_N is $d - Y_d + Z_{N-d}^{(j)}$, where d is the size of the first flipped family. As d grows by one this quantity moves by at most one, so it hits m at some d close to m unless Y or Z is large. (If no step of the family flips back and the rest of the tree has no flip, then $Y = Z = 0$ and the quantity is d itself, which hits m at $d = m$.) The law β_j of d is nearly flat near m, and this gives almost the full weight $\beta_j(m)$. Then we sum over j. The error e_1 collects the chance that Y or Z is large, the small slope of β_j, and the first flips that come too late to be treated this way.

Proof. We bound the inner sum of Lemma 3.3 from below for each early first flip, namely for $2 \le j \le J := \lfloor N/8 \rfloor$. The terms with $j > J$ are nonnegative and we drop them.

Step 1. One value of j. Fix $2 \le j \le J$ and put $\Gamma_d = d - Y_d + Z_{N-d}^{(j)}$ for $1 \le d \le N - j + 1$. The inner sum in Lemma 3.3 is $Q_j(m) = \mathbb{E}S_j$ with $S_j = \sum_d \beta_j(d)\mathbf{1}\{\Gamma_d = m\}$. Let $\mathcal{E}$ be the event $\{Z_{N-1}^{(j)} \le m-1,\ Y_{2m} \le m\}$. We claim that

$$S_j \ge \beta_j(m)\Big(\mathbf{1}_{\mathcal{E}} - \frac{2(j-2)}{N} Y_{2m}\Big).$$

Off $\mathcal{E}$ the right side is at most 0 and $S_j \ge 0$. So let $\mathcal{E}$ hold. As d grows by one, Y_d grows by 0 or 1 and $Z^{(j)}_{N-d}$ falls by 0 or 1. So every step of Γ lies in $\{-1,0,1\}$, and Γ cannot jump over m. It starts at $\Gamma_1 = 1 + Z^{(j)}_{N-1} \le m$ (note that $Y_1 = 0$). At $d^\circ = m + Y_{2m} \le 2m$ we have $\Gamma_{d^\circ} \ge d^\circ - Y_{d^\circ} \ge d^\circ - Y_{2m} = m$, since $Z^{(j)} \ge 0$ and Y is nondecreasing. (The index d° is allowed, because $2m + j \le N$.) So $\Gamma_d = m$ for some $d \le d^\circ$, and as β_j is nonincreasing, $S_j \ge \beta_j(d^\circ) = \beta_j(m + Y_{2m})$. It remains to compare $\beta_j(m+y)$ with $\beta_j(m)$ for $0 \le y \le m$. By (6),

$$\frac{\beta_j(m+y)}{\beta_j(m)} = \prod_{i=1}^{y}\Big(1 - \frac{j-2}{N-m-i}\Big) \ge 1 - \frac{2y(j-2)}{N}.$$

Here $N - m - i \ge N/2$, so each fraction is at most $\frac{2(j-2)}{N} \le \frac14$, and a product of factors $1-q_i$ with $0 \le q_i \le 1$ is at least $1 - \sum_i q_i$. This proves the claim. Now $\mathbf{1}_{\mathcal{E}} \ge 1 - \mathbf{1}\{Y_{2m} > m\} - \mathbf{1}\{Z^{(j)}_{N-1} \ge m\}$. Taking means in the claim gives

$$Q_j(m) \ge \beta_j(m)\big[1 - \mathbb{P}(Y_{2m} > m) - \mathbb{P}(Z^{(j)}_{N-1} \ge m)\big] - \frac{2(j-2)}{N}\,\mathbb{E}Y_{2m}\,\beta_j(m).$$

Step 2. The chance that Y or Z is large. Lemma B.2 gives

$$\mathbb{P}(Y_{2m} > m) \le 2\varepsilon H_{m+1}, \qquad \mathbb{P}(Z^{(j)}_{N-1} \ge m) \le \frac{xH_m}{m}, \qquad \mathbb{E}Y_{2m} \le 2\varepsilon m \log N.$$

None of these bounds depends on j. Put $p = 2\varepsilon H_{m+1} + xH_m/m$. If $p > 1$, then $e_1'(m) > 1$ and there is nothing to prove. So let $p \le 1$.

Step 3. Sum over j. By Lemma 3.3, $f_N(m) \ge \sum_{j=2}^{J} \mathbb{P}(V=j)\,Q_j(m)$. With Steps 1 and 2 this gives

$$f_N(m) \ge (1-p)\sum_{j=2}^{J}\mathbb{P}(V=j)\,\beta_j(m) - \frac{4\varepsilon m\log N}{N}\sum_{j=2}^{J}\mathbb{P}(V=j)\,(j-2)\,\beta_j(m).$$

In the second sum we use $\mathbb{P}(V=j) \le \varepsilon$ and the second sum in Lemma 2.4. So the second term is at most

$$\frac{4\varepsilon m\log N}{N}\cdot\frac{2x(N-m-1)}{m(m+1)(m+2)} \le \frac{x}{m(m+1)}\,8\varepsilon\log N.$$

The first sum is $\mathbb{P}(|\mathcal{T}_V| = m)$ without the terms $j > J$. So by Lemma B.3 it is at least $\frac{x}{m(m+1)}(1 - \frac{2x}{m+2}) - \varepsilon\sum_{j>J}\beta_j(m)$.

Step 4. The late first flips $j > J$. A family that starts after time $N/8$ rarely reaches size m. By (5), $\beta_j(m) \le \frac{j-1}{N-m} r^{j-1}$ with $r = e^{-a}$ and $a = \frac{m-1}{N-1} \le \frac14$. We sum this over $j - 1 \ge J$. With $\sum_{i\ge J} i r^i \le r^J\big(\frac{J}{1-r} + \frac{1}{(1-r)^2}\big)$, $1 - e^{-a} \ge a/2$ and $N - m \ge \frac34 N$ we get $\varepsilon\sum_{j>J}\beta_j(m) \le \frac{4\varepsilon}{3N}\big(\frac{2J}{a} + \frac{4}{a^2}\big)e^{-aJ}$. Since $\lfloor N/8\rfloor \ge \frac{N-1}{9}$ for $N \ge 64$, we have $aJ \ge \frac{m-1}{9}$. Also $J \le N/8$ and $a \ge \frac{m-1}{N}$. So

$$\varepsilon\sum_{j>J}\beta_j(m) \le x\Big(\frac{1}{3(m-1)} + \frac{16}{3(m-1)^2}\Big)e^{-(m-1)/9} \le \frac{x}{m(m+1)}\,20me^{-m/9}.$$

The last step uses $m+1 \le 3(m-1)$ and $m+1 \le 3(m-1)^2$, which hold for $m \ge 2$, and $e^{1/9} \le \frac98$, since $17 \cdot \frac98 \le 20$.

Step 5. Collect. For $0 \le p \le 1$ and $q \ge 0$ we have $(1-p)(1-q) \ge 1-p-q$. So Steps 3 and 4 give

$$f_N(m) \ge \frac{x}{m(m+1)}\Big(1 - p - \frac{2x}{m+2} - 20me^{-m/9} - 8\varepsilon \log N\Big),$$

which is the bound with $e_1'(m)$. For $e_1' \le e_1$ we use $\log(m+1) \ge \log 3 > 1$ and $\log N \ge \log 64 > 4$. So $H_m \le 1 + \log m \le 2\log(m+1)$, $H_{m+1} \le 1 + \log N \le \frac{5}{4}\log N$ and $\frac{2x}{m+2} \le \frac{2x\log(m+1)}{m}$. Hence $e_1'(m) \le \frac{4x\log(m+1)}{m} + \frac{21}{2}\varepsilon \log N + 20me^{-m/9} \le e_1(m)$. □

Lemma B.5 (mode bound)**.** *Let $N \ge 64$ and $x \ge 1$, and put $\bar{c} = \frac{1}{40}$ and $A_0 = 15400$. If $\varepsilon \log N \le \bar{c}$ and $804x\log(x+2) \le N$, then $m_h \le A_0 x \log^2(x+1)$. In particular this holds if $\varepsilon \log N \le \bar{c}$ and $A_0 x \log(x+2) \le N/4$.*

Proof. We pick a point ℓ of order $x \log x$ at which $f_N(\ell)$ is at least half of $\frac{x}{\ell(\ell+1)}$. Up to its mode f_N is nondecreasing. So if the mode were far to the right of ℓ, the bins from ℓ to the mode would carry more mass than the tail $\mathbb{P}(M_N \ge \ell)$ has.

Put $L_2 = \log(x+2) \ge \log 3 > 1$ and $\ell = \lceil 200xL_2 \rceil$, so that $200 \le \ell \le 201xL_2$ and $4\ell \le 804xL_2 \le N$. Since $\log N \ge 4$, we have $\varepsilon \le \frac{1}{160}$. We bound the five terms of $e_1'(\ell)$ in Proposition B.4. As $(1+\log s)/s$ decreases for $s \ge 1$, and $\log 200 \le 6$, $\log x \le L_2$ and $\log L_2 \le L_2$, we get $\frac{xH_\ell}{\ell} \le \frac{x(1+\log(200xL_2))}{200xL_2} \le \frac{7+2L_2}{200L_2} \le \frac{9}{200}$. Next $\frac{2x}{\ell+2} \le \frac{1}{100}$, $2\varepsilon H_{\ell+1} \le 2\varepsilon(1+\log N) \le \frac{1}{80} + \frac{1}{20}$ and $8\varepsilon \log N \le \frac{1}{5}$. Finally $se^{-s/9}$ decreases for $s \ge 9$, so $20\ell e^{-\ell/9} \le 4000e^{-200/9} < 10^{-5}$. The five terms add up to less than $\frac{1}{2}$, so $f_N(\ell) \ge \frac{x}{2\ell(\ell+1)}$. If $m_h \ge \ell$, Lemmas B.1 (with $a = \ell$, $b = m_h$) and B.2 give $(m_h - \ell + 1)f_N(\ell) \le \mathbb{P}(M_N \ge \ell) \le xH_\ell/\ell$. So $m_h \le (\ell+1)(1+2H_\ell)$ in every case. Here $\ell + 1 \le 202xL_2$, because $xL_2 \ge 1$. Also $H_\ell \le 1 + \log(201xL_2) \le 7 + 2L_2 \le 9L_2$, so $1 + 2H_\ell \le 19L_2$. Hence $m_h \le 3838\, xL_2^2$. Since $(x+1)^2 \ge x+2$, we have $L_2 \le 2\log(x+1)$, and so $m_h \le 15352\, x\log^2(x+1) \le A_0 x \log^2(x+1)$. The last claim holds because $804x\log(x+2) \le 4A_0 x \log(x+2)$. □

C Proofs for the bulk

We prove Theorems 3.8 and 3.9, Lemma 3.10 and Corollary 3.11. We keep the notation of Section 3 and of Appendix B. So $\mathcal{T}$ is the tree of Lemma 3.1, $\mathcal{T}_v$ is the family of v, and μ_v is the mark of v. By Lemma 2.4 and the discussion after Lemma 3.1, $|\mathcal{T}_v|$ has the law β_v, with mean N/v, and $\sum_{v=2}^N \beta_v(d) = N/(d(d+1))$ for $1 \le d \le N-1$. Let

$$\Psi_N(\theta) = \sum_{d=1}^{N-1} \frac{e^{i\theta d} - 1}{d(d+1)}, \qquad \Psi_\infty(\theta) = \sum_{d \ge 1} \frac{e^{i\theta d} - 1}{d(d+1)}.$$

The law CP_N has the characteristic function $\widehat{\mathrm{CP}}_N(\theta) = e^{x\Psi_N(\theta)}$.

C.1 Proof of Theorem 3.8

Here we assume only $N \ge 1$ and $0 < \varepsilon < 1$. Let $Z_+ = \sum_{v=2}^N \mu_v |\mathcal{T}_v|$ be the total size of the flipped families.

Step 1: $M_N = Z_+$ with high probability. Two families are nested or disjoint. Suppose that no marked vertex has a marked strict descendant. Then the marked families are disjoint, each of their vertices has exactly one mark on its path to the root, and the other vertices have none. So

$M_N = Z_+$ by Lemma 3.1. The number of pairs of a marked vertex and a marked strict descendant has mean $\varepsilon^2\,\mathbb{E}\sum_v(|\mathcal{T}_v|-1)$, since marks and tree are independent. Hence

$$\mathbb{P}(M_N \neq Z_+) \le \varepsilon^2 \sum_{v=2}^{N}\Big(\frac{N}{v}-1\Big) = \varepsilon^2(NH_N - 2N + 1).$$

Step 2. Given the tree, a product. Fix $\theta \in [-\pi,\pi]$. Given $\mathcal{T}$ the marks are independent, so $\mathbb{E}[e^{i\theta Z_+} \mid \mathcal{T}] = \prod_{v=2}^{N}(1+a_v)$ with $a_v = \varepsilon(e^{i\theta|\mathcal{T}_v|}-1)$. Here $|a_v| \le 2\varepsilon$, $\operatorname{Re} a_v \le 0$ and $|1+a_v| = |1-\varepsilon+\varepsilon e^{i\theta|\mathcal{T}_v|}| \le 1$. Put $W = \sum_v a_v$, so that $\operatorname{Re} W \le 0$ and $e^W = \prod_v e^{a_v}$ with $|e^{a_v}| \le 1$. For complex a with $\operatorname{Re} a \le 0$, Taylor's formula $e^a - 1 - a = a^2\int_0^1 (1-s)e^{sa}\,ds$ gives $|e^a - (1+a)| \le \frac{1}{2}|a|^2$. For complex numbers u_v and w_v of modulus at most 1 we have $|\prod_v u_v - \prod_v w_v| \le \sum_v |u_v - w_v|$, by changing one factor at a time. So

$$\Big|\prod_v (1+a_v) - e^W\Big| \le \sum_{v=2}^{N} \tfrac{1}{2}|a_v|^2 \le 2\varepsilon^2(N-1).$$

Step 3. Replace W by its mean. Write $W = \varepsilon F$ with $F = \sum_{v=2}^{N}(e^{i\theta|\mathcal{T}_v|}-1)$. Then $\mathbb{E}W = \varepsilon\sum_d(e^{i\theta d}-1)\sum_v \beta_v(d) = x\Psi_N(\theta)$, which is $\log \widehat{\mathrm{CP}}_N(\theta)$. Put $\delta = W - \mathbb{E}W$. Taylor's formula for $s \mapsto e^{\mathbb{E}W + s\delta}$ on $[0,1]$ gives

$$e^W - e^{\mathbb{E}W} - e^{\mathbb{E}W}\delta = \delta^2 \int_0^1 (1-s)\, e^{\mathbb{E}W + s\delta}\, ds.$$

The exponent $\mathbb{E}W + s\delta = (1-s)\mathbb{E}W + sW$ has real part at most 0, so the right side has modulus at most $\frac{1}{2}|\delta|^2$. As $\mathbb{E}\delta = 0$, we get $|\mathbb{E}e^W - e^{\mathbb{E}W}| \le \frac{1}{2}\varepsilon^2 \operatorname{Var} F$, where $\operatorname{Var} F = \mathbb{E}|F - \mathbb{E}F|^2$.

Step 4. The variance of F. The variable F is a function of the independent parents $U_1,\dots,U_{N-1}$. Let $F^{(k)}$ be F with U_k replaced by an independent copy U_k'. The Efron–Stein inequality [27] says that $\operatorname{Var} F \le \frac{1}{2}\sum_k \mathbb{E}|F - F^{(k)}|^2$. It holds for every square-integrable function of independent variables [32], and it holds for the complex F because both sides are the sums of those for $\operatorname{Re} F$ and $\operatorname{Im} F$.

Replacing U_k by U_k' moves the family of $k+1$ from U_k to U_k'. So $|\mathcal{T}_v|$ changes only for vertices $v \ge 2$ on the path from U_k to the root or on the path from U_k' to the root. These paths do not depend on the choice at step k, since $U_k, U_k' \le k$. Let $\mathrm{dp}(u)$ be the depth of u, namely the number of edges between u and the root. There are at most $\mathrm{dp}(U_k) + \mathrm{dp}(U_k')$ such vertices, and each term of F moves by at most 2. Hence $\mathbb{E}|F - F^{(k)}|^2 \le 8\,\mathbb{E}[\mathrm{dp}(U_k)^2 + \mathrm{dp}(U_k')^2] = 16\,\mathbb{E}\,\mathrm{dp}(U_k)^2$.

Since $\mathrm{dp}(u+1) = 1 + \mathrm{dp}(U_u)$, the generating function $g_u(s) = \mathbb{E}s^{\mathrm{dp}(u)}$ satisfies $g_1 = 1$ and $u\,g_{u+1}(s) = s\sum_{j\le u} g_j(s)$. Subtracting the same identity for $u-1$ gives $u\,g_{u+1}(s) = (u-1+s)\,g_u(s)$ for $u \ge 2$, and $g_2(s) = s$. So $g_u(s) = \prod_{i=1}^{u-1}\frac{i-1+s}{i}$. So $\mathrm{dp}(u)$ is a sum of independent Bernoulli$(1/i)$ variables with $1 \le i < u$, and $\mathbb{E}\,\mathrm{dp}(u)^2 \le H_{u-1}^2 + H_{u-1}$. The parent U_k is uniform on $\{1,\dots,k\}$ and independent of the depths of $1,\dots,k$, so $\mathbb{E}\,\mathrm{dp}(U_k)^2 \le H_{k-1}^2 + H_{k-1}$. We get

$$\operatorname{Var} F \le \frac{1}{2}\sum_{k=1}^{N-1} 16\big(H_{k-1}^2 + H_{k-1}\big) = 8\sum_{k=0}^{N-2}\big(H_k^2 + H_k\big),$$

and the bound of Step 3 is at most $4\varepsilon^2\sum_{k=0}^{N-2}(H_k^2 + H_k)$.

Step 5: Fourier inversion. For every integer m,

$$\mathbb{P}(Z_+ = m) - \mathrm{CP}_N(m) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-im\theta}\big(\mathbb{E}e^{i\theta Z_+} - \widehat{\mathrm{CP}}_N(\theta)\big)\,d\theta,$$

so the left side is at most $\sup_\theta |\mathbb{E}e^{i\theta Z_+} - \widehat{\mathrm{CP}}_N(\theta)|$ in absolute value. By Steps 2 to 4 this is at most $2\varepsilon^2(N-1)+4\varepsilon^2\sum_{k=0}^{N-2}(H_k^2+H_k)$. Finally $|f_N(m)-\mathbb{P}(Z_+=m)| \le \mathbb{P}(M_N \ne Z_+)$, and Step 1 bounds it. The three bounds add up to δ_Σ. For $\delta_\Sigma \le \delta_B$ we compare term by term. Since $H_N \le 1+\log N$, we have $\varepsilon^2(NH_N - 2N + 1) \le \varepsilon^2 N \log N = x^2 \log N/N$. Next $2\varepsilon^2(N-1) \le 4\varepsilon^2 N \le \frac{4x^2}{N} e^{4x^2/N}$. Finally $H_k \le H_N$ for $k \le N-2$, so the sum is at most $4\varepsilon^2 N(H_N^2 + H_N) = 4x^2(H_N^2+H_N)/N$. This proves Theorem 3.8. □

C.2 Proof of Theorem 3.9

We write $B(t) = -it\log|t| - \frac{\pi}{2}|t|$ for real $t \ne 0$, and $B(0) = 0$. For $t > 0$ this is the value of $u \log u$ at $u = -it$, with the principal logarithm, and $B(-t) = \overline{B(t)}$. We need three facts about Ψ_N and Ψ_∞.

Fact 1. For real θ, $|\Psi_N(\theta) - \Psi_\infty(\theta)| \le \sum_{d \ge N} \frac{2}{d(d+1)} = \frac{2}{N}$.

Fact 2. For $0 < |\theta| \le \pi$ we have $\Psi_\infty(\theta) = -i\theta\log(-i\theta) + R(\theta)$ with $|R(\theta)| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big)$. The reason is that Ψ_∞ has a closed form in which θ enters only through $1-e^{i\theta}$, and $1-e^{i\theta}$ is close to $-i\theta$ for small θ. The term R is what this replacement changes. We prove it in the next four paragraphs.

We start with the closed form. For $0 < |s| \le 1$, $s \ne 1$,

$$\sum_{d\ge 1} \frac{s^d - 1}{d(d+1)} = \chi(s) := \frac{(1-s)\log(1-s)}{s},$$

because $\frac{1}{d(d+1)} = \frac{1}{d} - \frac{1}{d+1}$, $\sum_d s^d/d = -\log(1-s)$, $\sum_d s^d/(d+1) = (-\log(1-s) - s)/s$ and $\sum_d \frac{1}{d(d+1)} = 1$. So $\Psi_\infty(\theta) = \chi(e^{i\theta})$.

Next we compare $1-e^{i\theta}$ with $-i\theta$. Write $1 - e^{i\theta} = (-i\theta)\,\mathsf{u}$ with $\mathsf{u} = \mathrm{sinc}(\theta/2)\,e^{i\theta/2}$, where $\mathrm{sinc}\,y = \sin y/y$ is positive for $|y| \le \frac{\pi}{2}$. So u is close to 1 when θ is small. The argument of $-i\theta$ is $\mp\frac{\pi}{2}$ for $\pm\theta > 0$ and that of u is $\theta/2$. They add up to a number in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\log(1-e^{i\theta}) = \log(-i\theta) + \log\mathsf{u}$ for principal logarithms.

Now we get a formula for R. By the two steps above, $\chi(e^{i\theta}) = (-i\theta)\,\mathsf{u}e^{-i\theta}\big(\log(-i\theta) + \log\mathsf{u}\big)$. Subtracting $-i\theta\log(-i\theta)$ gives

$$R(\theta) = -i\theta\log(-i\theta)\big(\mathsf{u}e^{-i\theta} - 1\big) - i\theta\,\mathsf{u}e^{-i\theta}\log\mathsf{u}.$$

It remains to bound the three quantities in this formula. We show that

$$|\mathsf{u}e^{-i\theta} - 1| \le 0.631\,|\theta|, \qquad |\log\mathsf{u}| \le 0.53\,|\theta|, \qquad |\log(-i\theta)| \le |\log|\theta|| + \frac{\pi}{2}.$$

For the first, put $y = \theta/2$. Then $|\mathsf{u}e^{-i\theta} - 1| = |\mathrm{sinc}\,y - e^{iy}| \le (1 - \mathrm{sinc}\,y) + |1 - e^{iy}| \le \frac{y^2}{6} + |y| \le (\frac{1}{2} + \frac{\pi}{24})|\theta|$, where the last step uses $|\theta| \le \pi$, and $\frac{1}{2} + \frac{\pi}{24} \le 0.631$. For the second, $|\log\mathsf{u}|^2 = \log^2\mathrm{sinc}(\theta/2) + \theta^2/4$, and $-\log\mathrm{sinc}\,y = \sum_{k\ge1} \frac{\zeta(2k)}{k\pi^{2k}} y^{2k}$ has positive coefficients. So $|\log\mathrm{sinc}(\theta/2)|/|\theta|$ increases in $|\theta|$, and its value at $|\theta| = \pi$ gives $|\log\mathsf{u}| \le \big(\frac{1}{4} + \frac{\log^2(\pi/2)}{\pi^2}\big)^{1/2}|\theta| \le 0.53\,|\theta|$. The third holds because $-i\theta$ has modulus $|\theta|$ and argument $\pm\frac{\pi}{2}$. Since $|\mathsf{u}e^{-i\theta}| \le 1$, the formula for R now gives

$$|R(\theta)| \le |\theta|\Big(|\log|\theta|| + \frac{\pi}{2}\Big)\cdot 0.631\,|\theta| + |\theta|\cdot 0.53\,|\theta| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big),$$

as $0.631\cdot\frac{\pi}{2} + 0.53 \le 1.53$.

Fact 3. For $|\theta| \le \pi$, $-\operatorname{Re}\Psi_\infty(\theta) = \sum_d \frac{1-\cos\theta d}{d(d+1)} \ge \frac{|\theta|}{2\pi}$. We may take $\theta \ne 0$. Keep only the terms with $d \le D := \lfloor \pi/|\theta| \rfloor$ and use $1-\cos v \ge 2v^2/\pi^2$ for $|v| \le \pi$. This gives at least $\frac{2\theta^2}{\pi^2}\sum_{d\le D}\frac{d}{d+1} \ge \frac{\theta^2 D}{\pi^2} \ge \frac{|\theta|}{2\pi}$, since $D \ge \frac{\pi}{2|\theta|}$.

The function e^B is a characteristic function of class L. Let S_x have the law CP_∞ with parameter x, namely the law of $\sum_{d\ge1} d\,\mathcal{N}_d$. Its characteristic function is $e^{x\Psi_\infty(\theta)}$. By Fact 2, for $t \ne 0$ and $x \ge \max(1, |t|/\pi)$,

$$x\Psi_\infty(t/x) - it\log x = B(t) + xR(t/x), \qquad |xR(t/x)| \le \frac{t^2}{x}\bigl(0.631(|\log|t|| + \log x) + 1.53\bigr). \quad (7)$$

So the characteristic function of $(S_x - x\log x)/x$ tends to $e^{B(t)}$ as $x \to \infty$, for every real t. The limit is continuous at 0, so by Lévy's continuity theorem it is the characteristic function of a probability law. It is integrable, so this law has the bounded continuous density f_L of Theorem 3.9. For $0 < c < 1$ and real t we have $B(ct) = cB(t) - ict\log c$, and hence

$$e^{B(t)} = e^{B(ct)} \cdot e^{B((1-c)t)}\, e^{it[(1-c)\log(1-c) + c\log c]}.$$

The last two factors form the characteristic function of a scaled and shifted copy of the law. So the law is of class L (self-decomposable), and f_L is unimodal by [28].

Proof of (a). By Fourier inversion on $\mathbb{Z}$ and the substitution $\theta = t/x$,

$$x\,\mathrm{CP}_N(m) = \frac{1}{2\pi}\int_{-\pi x}^{\pi x} e^{-i\lambda_m t} e^{A(t)}\,dt, \qquad A(t) = x\Psi_N(t/x) - it\log x.$$

Let $|t| \le \pi x$. By (7) and Fact 1,

$$|A(t) - B(t)| \le \frac{t^2}{x}\bigl(0.631(|\log|t|| + \log x) + 1.53\bigr) + \frac{2x}{N},$$

and by Facts 1 and 3, $\max(\operatorname{Re}A(t), \operatorname{Re}B(t)) \le -\frac{|t|}{2\pi} + \frac{2x}{N}$. Now $|e^A - e^B| \le |A-B|\, e^{\max(\operatorname{Re}A, \operatorname{Re}B)}$, and $\frac{2x}{N} \le \frac12$ because $N \ge 4x$. So for $|t| \le \pi x$

$$|e^{A(t)} - e^{B(t)}| \le e^{1/2} e^{-|t|/2\pi}\Bigl[\frac{t^2}{x}\bigl(0.631(|\log|t|| + \log x) + 1.53\bigr) + \frac{2x}{N}\Bigr].$$

For $s \ge 1$ the tangent of the logarithm at 16 gives $\log s \le s/16 + \log 16 - 1$, and for $0 < s < 1$ we have $s|\log s| \le 1/e$. So $s^2|\log s| \le s^3/16 + (4\log 2 - 1)s^2 + s/e$ for every $s \ge 0$. We insert this with $s = |t|$ and integrate, using $\int_0^\infty t^k e^{-t/2\pi}\,dt = k!\,(2\pi)^{k+1}$. This gives

$$\frac{1}{2\pi}\int_{-\pi x}^{\pi x} |e^{A(t)} - e^{B(t)}|\,dt \le e^{1/2}\Bigl[\frac{K_0 + 16\pi^2 \cdot 0.631\log x}{x} + \frac{4x}{N}\Bigr],$$

where $K_0 = 6\pi^3 \cdot 0.631 + 16\pi^2\bigl(0.631(4\log 2 - 1) + 1.53\bigr) + 4\pi \cdot 0.631/e \le 538.6$. Beyond πx we use $|e^{B(t)}| = e^{-\pi|t|/2}$, which gives $\frac{1}{2\pi}\int_{|t|>\pi x}|e^{B(t)}|\,dt = \frac{2}{\pi^2}e^{-\pi^2 x/2} \le \frac{4}{\pi^4 x} \le \frac{0.042}{x}$, since $e^{-y} \le 1/y$ for $y > 0$. Since $f_L(\lambda_m) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-i\lambda_m t} e^{B(t)}\,dt$, the error is at most the sum of the two integrals. With $e^{1/2} \le 1.6488$ and $16\pi^2 \cdot 0.631 \le 99.66$ this sum is at most

$$\frac{1.6488 \cdot 538.6 + 0.042 + 1.6488 \cdot 99.66\log x}{x} + \frac{6.5952\,x}{N} \le \frac{889 + 165\log x}{x} + \frac{6.6\,x}{N}.$$

The second bound in (a) holds because $165 \le 889$. This proves (a). Fourier inversion on $\mathbb{Z}$ holds at every integer m, and $\mathrm{CP}_N(m) = 0$ for $m < 0$, so the proof covers every integer m.

Proof of (b). Since $\varepsilon \le \frac{1}{4}$, we have $N \ge 4x$. By Theorem 3.8, $|x f_N(m) - x\,\mathrm{CP}_N(m)| \le x\delta_\Sigma \le x\delta_B$. Add (a).

Proof of (c). We drop the index n. Let $x \to \infty$ and $x^3 \log^2 N/N \to 0$. Then $N \ge x \to \infty$, so $\log N \ge 1$ for large n, and $x/N \to 0$. Hence for large n we have $\varepsilon \le \frac{1}{4}$ and $x \ge 1$, and (b) applies. Its bound tends to 0, since $x\delta_B = O(x^3 \log^2 N/N)$. Only large n matter for convergence in law. Let U be uniform on $[0,1)$ and independent of M_N. Then $(M_N + U - x\log x)/x$ has the density q_N with $q_N(\lambda) = x f_N(m)$ for $\lambda_m \le \lambda < \lambda_m + \frac{1}{x}$. On a compact set of λ the relevant m are nonnegative once x is large, and $|\lambda - \lambda_m| < \frac{1}{x}$. As f_L is uniformly continuous, $q_N \to f_L$ uniformly on compact sets. By Scheffé's lemma $q_N \to f_L$ in L^1, and since $U/x \to 0$ this gives the convergence in law. At the crossing $x_N \sim 2\log N$ by Theorem 2.7, so both conditions hold. □

C.3 Proof of Lemma 3.10

Step 1. A real integral. We claim that for every real λ and $k \ge 0$,

$$f_L^{(k)}(\lambda) = \frac{1}{\pi} \int_0^\infty (-s)^k\, e^{-s\log s - \lambda s} \sin(\pi s)\, ds. \tag{8}$$

For $k = 0$ this is [20, Eq. (4.3)], and we include a short proof. Since $B(-t) = \overline{B(t)}$, the definition of f_L gives $f_L(\lambda) = \frac{1}{\pi}\,\mathrm{Re}\int_0^\infty \Phi(-it)\,dt$ with $\Phi(u) = \exp(u\log u + \lambda u)$. The function Φ is holomorphic off the negative real axis. Put $u = \rho e^{i\varphi}$ with $-\pi < \varphi \le -\frac{\pi}{2}$. Then $\mathrm{Re}(u\log u) = \rho(\log\rho\cos\varphi - \varphi\sin\varphi)$, where $\cos\varphi \le 0$ and $\varphi\sin\varphi \ge 0$. For $\log\rho \ge |\lambda| + 1$ this gives $\mathrm{Re}(u\log u + \lambda u) \le -\rho(|\cos\varphi| + \varphi\sin\varphi) \le -\rho$, because $|\cos\varphi| + \varphi\sin\varphi \ge 1$ on $[-\pi, -\frac{\pi}{2}]$ (for $\psi = \varphi + \pi \in [0, \frac{\pi}{2}]$ it equals $\cos\psi + (\pi - \psi)\sin\psi \ge \cos\psi + \sin\psi \ge 1$). Also Φ is bounded near 0. So by Cauchy's theorem we may turn the ray $u = -it$ to the lower side of the negative real axis, where $u = -s$ and $\log u = \log s - i\pi$. There $u\log u = -s\log s + i\pi s$, and $dt = i\,du = -i\,ds$. Hence

$$f_L(\lambda) = \frac{1}{\pi}\,\mathrm{Re}\Big[-i\int_0^\infty e^{-s\log s - \lambda s}\, e^{i\pi s}\, ds\Big] = \frac{1}{\pi}\int_0^\infty e^{-s\log s - \lambda s}\sin(\pi s)\, ds.$$

The integrand decays faster than every exponential, so we may differentiate under the integral sign. This proves (8). Also f_L'' is bounded on the real line, because $t^2 e^{B(t)}$ is integrable and we may differentiate the Fourier formula for f_L twice.

Step 2. Certified enclosures. Let $-0.223 \le \lambda \le 0$ and $k \in \{1, 2\}$, and split the integral in (8) at $\delta = 2^{-20}$ and at 30. On $[0, \delta]$ we have $-s\log s \le 1/e$, $-\lambda s \le 0.223\,s$ and $|\sin\pi s| \le \pi s$, so this part is at most $\pi e^{1/e + 0.223\delta}\delta^{k+2}/(k+2)$ in absolute value. On $[30, \infty)$ we have $s\log s + \lambda s \ge cs$ with $c = \log 30 - 0.223 > 3$, so this part is at most $\int_{30}^\infty s^2 e^{-cs}\, ds \le 30^2 e^{-30c}$. Both are below 10^{-17}. On $[\delta, 30]$ the integrand is holomorphic in a neighborhood of the path, and we enclose the integral with the rigorous integration algorithm of [29], in ball arithmetic with 128 bits. The parameter λ enters as a ball, so one run encloses the integral for every λ in an interval. The script `certify_elephant.py` does this and reports

$$f_L'(-0.2229) \in [9.232 \cdot 10^{-6}, 9.234 \cdot 10^{-6}], \qquad f_L'(-0.2227) \in [-6.548 \cdot 10^{-6}, -6.546 \cdot 10^{-6}],$$

and $f_L''(\lambda) \in [-0.0792, -0.0786]$ for every $\lambda \in [-0.2229, -0.2227]$. In the same way it finds that f_L' is positive at -0.22278299 and negative at -0.22278297. The two enclosures are $6.8984026 \cdot 10^{-10}$ and $-8.8809315 \cdot 10^{-10}$ with radii below 10^{-17}, and these are the smallest margins in the proof.

Step 3. The mode. Write $[a, b] = [-0.2229, -0.2227]$. By Step 2, f_L' is strictly decreasing on $[a, b]$ with $f_L'(a) > 0 > f_L'(b)$. So f_L' has exactly one zero λ_{L} in $[a, b]$, and f_L is strictly increasing

on $[a, \lambda_{\mathrm{L}}]$ and strictly decreasing on $[\lambda_{\mathrm{L}}, b]$. Since f_L is unimodal, there is a point λ_* such that f_L is nondecreasing on $(-\infty, \lambda_*]$ and nonincreasing on $[\lambda_*, \infty)$. If $\lambda_* > \lambda_{\mathrm{L}}$, then f_L would be nondecreasing just to the right of λ_{L}, which is false. In the same way $\lambda_* < \lambda_{\mathrm{L}}$ is impossible. So $\lambda_* = \lambda_{\mathrm{L}}$, and f_L is nondecreasing on $(-\infty, \lambda_{\mathrm{L}}]$ and nonincreasing on $[\lambda_{\mathrm{L}}, \infty)$. By Step 2 again, $\lambda_{\mathrm{L}} \in (-0.22278299, -0.22278297)$. This proves (a) and (b).

Step 4. The constant κ_L. Let $\delta_0 = \min(\lambda_{\mathrm{L}} - a, b - \lambda_{\mathrm{L}}) > 0$. For $|\lambda - \lambda_{\mathrm{L}}| \le \delta_0$, Taylor's formula with $f_L'(\lambda_{\mathrm{L}}) = 0$ and $f_L'' \le -0.078$ on $[a, b]$ gives $f_L(\lambda_{\mathrm{L}}) - f_L(\lambda) \ge 0.039(\lambda - \lambda_{\mathrm{L}})^2$. For $|\lambda - \lambda_{\mathrm{L}}| > \delta_0$ the monotonicity of Step 3 gives $f_L(\lambda) \le f_L(\lambda_{\mathrm{L}} \pm \delta_0) \le f_L(\lambda_{\mathrm{L}}) - 0.039\,\delta_0^2$. Since $\delta_0 < 1$, the bound in (c) holds with $0.039\,\delta_0^2$ in place of κ_L. By (a), $\lambda_{\mathrm{L}} - a > 0.2229 - 0.22278299 > 1.17 \cdot 10^{-4}$ and $b - \lambda_{\mathrm{L}} > 0.22278297 - 0.2227 = 8.297 \cdot 10^{-5}$, so $0.039\,\delta_0^2 > 0.039 \cdot (8.297 \cdot 10^{-5})^2 > 2.68 \cdot 10^{-10}$. So (c) holds with $\kappa_L = 2.6 \cdot 10^{-10}$. □

C.4 Proof of Corollary 3.11

The method is to compare f_N at a mode with f_N at the integer m' nearest to the predicted place $x \log x + \lambda_{\mathrm{L}} x$. By Theorem 3.9(b) both values are close to values of f_L, and f_L falls off quadratically away from λ_{L}. So a mode cannot be far from m'.

We first prove (a). So we assume (H), $E' < \kappa_L/2$ and $m_h \le N/4$, with E_1 and E' as in the statement.

Let m' be the integer nearest to $x \log x + \lambda_{\mathrm{L}} x$. This number increases in $x \ge 1$ and is at least $\lambda_{\mathrm{L}} > -\frac{1}{2}$, so $m' \ge 0$. Also $|\lambda_{m'} - \lambda_{\mathrm{L}}| \le \frac{1}{2x}$, and since $f_L'(\lambda_{\mathrm{L}}) = 0$, $f_L(\lambda_{m'}) \ge f_L(\lambda_{\mathrm{L}}) - \|f_L''\|_\infty/(8x^2)$. Let $\tilde{m}$ be a mode of f_N or $\tilde{m} = m^*$. We claim that $f_N(\tilde{m}) \ge f_N(m') - 16xH_N/N^2$. For a mode of f_N this is clear. For m^* we have $f_N(m^*) + f_N(N - m^*) = 2P_N(N - m^*) \ge 2P_N(N - m') \ge f_N(m')$. Here $N - m^* \ge \lceil N/2 \rceil \ge \lceil N/4 \rceil \ge m_h$. So f_N is nonincreasing on $\{\lceil N/4 \rceil, \dots, N\}$ by Lemma 3.4. There are at least $N/4$ integers from $\lceil N/4 \rceil$ to $\lceil N/2 \rceil$. So Lemma B.1 (with $a = \lceil N/4 \rceil \ge m_h$ and $b = \lceil N/2 \rceil$) gives the second inequality below, and Lemma B.2 gives the third,

$$f_N(N - m^*) \le f_N(\lceil N/2 \rceil) \le \frac{\mathbb{P}(M_N \ge \lceil N/4 \rceil)}{N/4} \le \frac{16xH_N}{N^2}.$$

Now Theorem 3.9(b), used at $\tilde{m}$ and at m', gives

$$f_L(\lambda_{\tilde{m}}) \ge x f_N(\tilde{m}) - E_1 \ge x f_N(m') - \frac{16x^2 H_N}{N^2} - E_1 \ge f_L(\lambda_{m'}) - 2E_1 - \frac{16x^2 H_N}{N^2} \ge f_L(\lambda_{\mathrm{L}}) - 2E'.$$

Since $2E' < \kappa_L$, Lemma 3.10(c) gives $(\lambda_{\tilde{m}} - \lambda_{\mathrm{L}})^2 \le 2E'/\kappa_L$, which is the claim.

Now we prove (b). Let N be large and $\log N \le x \le 2 \log N$. Then $x\delta_B$, x/N, $16x^2 H_N/N^2$ and $\|f_L''\|_\infty/(8x^2)$ are all at most $\log x/x$, because each is $O(\log^5 N/N)$ or $O(1/\log^2 N)$. So $E' \le C_1 \log x/x$ with an absolute constant C_1. As $\log x/x \le \log\log N/\log N$, for large N we have $E' < \kappa_L/2$. Also (H) holds, $\varepsilon \log N \le 2\log^2 N/N \le \frac{1}{40}$ and $804x \log(x+2) \le N$, so Lemma B.5 gives $m_h \le A_0 x \log^2(x+1) \le N/4$. Hence by (a) $|m^* - x\log x - \lambda_{\mathrm{L}} x| \le x\sqrt{2E'/\kappa_L} \le K\sqrt{x \log x}$ with $K = \sqrt{2C_1/\kappa_L}$, and the same holds for every mode of f_N. At the crossing $x_N = 2\log N - \log\log N + O(1)$ for even N, by Theorem 2.7(b). So $\log N \le x_N \le 2 \log N$ for large even N. □

D Proofs for the second difference

We keep the notation of Section 3. This appendix proves the results of Section 3.5 on the signs of the second difference.

D.1 Proof of Lemma 3.16

We insert the numbers η_i into ζ one at a time. If $\eta_i = 0$, the sign pattern does not change. If $\eta_i > 0$, then $\zeta_i > 0$ or $\zeta_{i+1} > 0$, so η_i is placed next to an entry of its own sign, and the pattern does not change. The case $\eta_i < 0$ is the same. This proves the first claim. Deleting entries from a list replaces its sign pattern by a subsequence, and η is obtained from w by deleting the entries ζ_i. This proves the second claim. □

D.2 Proof of Lemma 3.17

Put $\xi_t = P_t(1)/P_t(0)$. By (1), $P_{t+1}(0) = (1-\varepsilon)P_t(0)$ and $P_{t+1}(1) = (1-\varepsilon-\frac{c}{t})P_t(1) + \varepsilon P_t(0)$. So

$$\xi_{t+1} - \xi_t = \frac{1}{1-\varepsilon}\Big(\varepsilon - \frac{c\xi_t}{t}\Big), \qquad \xi_2 = \frac{2\varepsilon}{1-\varepsilon}.$$

We claim that $\xi_t < \varepsilon t/c$ for $t \geq 2$. For $t = 2$ this holds because $c < 1-\varepsilon$. If it holds for t, then $0 < \xi_{t+1} - \xi_t \leq \frac{\varepsilon}{1-\varepsilon} < \frac{\varepsilon}{c}$, and it holds for $t+1$. So ξ_t is strictly increasing for $t \geq 2$, and $\xi_t < 1$ whenever $t \leq c/\varepsilon$.

Now (2) with $a = 1$, and $D_t(0) = P_t(1) - 2P_t(0)$, give for $t \geq 2$

$$s_{t+1} = \big(1 - r_t(2)\big)s_t + \sigma_t, \qquad \sigma_t = \Big(\varepsilon - \frac{c}{t}\Big)(\xi_t - 2)\, P_t(0).$$

Here $1 - r_t(2) \geq 1 - \varepsilon - c = \varepsilon > 0$. If $t < c/\varepsilon$, both factors of σ_t are negative and $\sigma_t > 0$. If $t > c/\varepsilon$, the first factor is positive, and the second is first negative and then positive, because ξ_t increases. So $\mathrm{sp}(\sigma_2, \sigma_3, \dots) \preceq \Pi_3$. Let $\Phi_t = \prod_{u=2}^{t-1}(1 - r_u(2)) > 0$. Then

$$\frac{s_t}{\Phi_t} = s_2 + \sum_{u=2}^{t-1} \frac{\sigma_u}{\Phi_{u+1}}, \qquad s_2 = 1 - 3\varepsilon.$$

If $\varepsilon < \frac{1}{3}$, the terms $s_2, \sigma_2, \sigma_3, \dots$ are first nonnegative, then nonpositive, then nonnegative. Their partial sums are nonnegative and nondecreasing in the first stretch, nonincreasing in the second, and nondecreasing in the third. So the sign pattern of the partial sums is a subsequence of Π_3. If $\varepsilon \geq \frac{1}{3}$, then $c/\varepsilon \leq 1$, so the terms are first nonpositive and then nonnegative, and the sign pattern of the partial sums is a subsequence of $(-,+)$. This proves the first claim.

For the second claim let $0 \leq \varepsilon \leq 1$. The proofs of (1) and (2) use only that step $t+1$ is $+1$ with probability $r_t(a)$ given $A_t = a$, so both hold for this ε. We compute $P_3(0) = \frac{1}{2}(1-\varepsilon)^2$ and $P_3(1) = P_3(2) = \frac{1}{2} - P_3(0)$, so $s_3 = P_3(0) - P_3(1) = (1-\varepsilon)^2 - \frac{1}{2}$. Suppose that $s_3 = 0$. Then $\varepsilon = 1 - 2^{-1/2}$ and $P_3(0) = P_3(1) = \frac{1}{4}$. By (2) with $t = 3$ and $a = 1$, $s_4 = r_3(-1)D_3(0) = (\varepsilon - \frac{c}{3})(P_3(1) - 2P_3(0)) = -\frac{5\varepsilon - 1}{12}$. This is not 0, because $\varepsilon \neq \frac{1}{5}$. (With P_t defined by (1), the same computation works for every real ε, because the other root $1 + 2^{-1/2}$ of s_3 is not $\frac{1}{5}$ either. We need only $0 < \varepsilon < \frac{1}{2}$.) □

D.3 Proof of Theorem 3.18

For $3 \leq t \leq N$ let $\mathcal{Z}_t$ be the list

$$\mathcal{Z}_t = \big(s_N, s_{N-1}, \dots, s_{t+1},\ s_t, D_t(2), D_t(3), \dots, D_t(t-2), s_t,\ s_{t+1}, \dots, s_{N-1}, s_N\big).$$

It has $2N - t - 1$ entries. Its middle part is $(D_t(1), D_t(2), \dots, D_t(t-1))$, because $D_t(t-1) = D_t(1) = s_t$ by symmetry. The list $\mathcal{Z}_3$ is $(s_N, \dots, s_4, s_3, s_3, s_4, \dots, s_N)$, and $\mathcal{Z}_N$ is the list in the statement.

We check that $\mathcal{Z}_{t+1}$ is obtained from $\mathcal{Z}_t$ as in Lemma 3.16, for $3 \le t \le N-1$. The first $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_N, \dots, s_{t+1}$, and they are copies of the first $N-t$ entries of $\mathcal{Z}_t$. So $A_i = 1$ and $B_i = 0$ there. The next $t-2$ entries are $D_{t+1}(b)$ for $2 \le b \le t-1$. By (2),

$$D_{t+1}(b) = r_t(b-2)\, D_t(b-1) + \big(1 - r_t(b+1)\big) D_t(b),$$

and $D_t(b-1)$ and $D_t(b)$ are neighbors in the middle part of $\mathcal{Z}_t$, at the right places. The coefficients satisfy $r_t(b-2) \ge \varepsilon$ and $1 - r_t(b+1) \ge 1 - \varepsilon - c = \varepsilon$, since $2 \le b \le t-1$. The last $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_{t+1}, \dots, s_N$, and each is a copy of the entry of $\mathcal{Z}_t$ one place to its right. So $A_i = 0$ and $B_i = 1$ there.

By Lemma 3.16, $\mathrm{sp}(\mathcal{Z}_N) \preceq \mathrm{sp}(\mathcal{Z}_{N-1}) \preceq \cdots \preceq \mathrm{sp}(\mathcal{Z}_3)$. By Lemma 3.17 the sign pattern p of $(s_3, \dots, s_N)$ is a subsequence of Π_3, so it is one of the lists $()$, $(+)$, $(-)$, $(+,-)$, $(-,+)$ and $(+,-,+)$. The sign pattern of $\mathcal{Z}_3$ is p reversed, followed by p, with equal neighbors merged. In the six cases this is $()$, $(+)$, $(-)$, $(-,+,-)$, $(+,-,+)$ and $(+,-,+,-,+)$, and each is a subsequence of Π_5.

For f_N we argue in the same way with $D_t^f(m) = f_t(m+1) - 2f_t(m) + f_t(m-1)$ and the lists

$$\mathcal{Z}_t^f = \big(s_N^f, \dots, s_{t+1}^f,\ s_t^f, D_t^f(2), \dots, D_t^f(t-1),\ d_t, d_{t+1}, \dots, d_{N-1}\big), \qquad 2 \le t \le N,$$

where $s_t^f = D_t^f(1)$ and $d_u = (1 - r_u(u+1))D_u^f(u) = (\varepsilon - \frac{c}{u}) f_u(u-1)$. The entry $D_{t+1}^f(t)$ of $\mathcal{Z}_{t+1}^f$ equals $r_t(t-2)D_t^f(t-1) + d_t$ by (2), and the other entries are as before. The proof of Lemma 3.17 applies to s_t^f without change, with $\xi_t = f_t(1)/f_t(0)$, $\xi_2 = \frac{\varepsilon}{1-\varepsilon}$ and $s_2^f = 1 - 3\varepsilon$. The numbers d_u are negative for $u < c/\varepsilon$ and positive for $u > c/\varepsilon$. So $\mathrm{sp}(\mathcal{Z}_2^f)$ is a subsequence of $(+,-,+)$ followed by $(-,+)$, and $\mathcal{Z}_N^f = (D_N^f(1), \dots, D_N^f(N-1))$. As before $\mathrm{sp}(\mathcal{Z}_N^f) \preceq \mathrm{sp}(\mathcal{Z}_2^f)$, and a subsequence of $(+,-,+)$ followed by $(-,+)$ is a subsequence of Π_5 once equal neighbors are merged. This argument needs only $N \ge 2$, and for $N = 2$ the lists $\mathcal{Z}_2^f$ and $\mathcal{Z}_N^f$ are the same. □

D.4 Proof of Theorem 3.19

By symmetry $D_N(N - a_1) > 0 > D_N(N - a_0)$. Suppose that $D_N(a) \le 0$ for some a with $a_1 < a < N - a_1$.

The case $D_N(a) < 0$. Then the entries of $\mathcal{Z}_N$ at $a_0 < a_1 < a < N - a_1 < N - a_0$ have the signs $-,+,-,+,-$. So $(-,+,-,+,-) \preceq \mathrm{sp}(\mathcal{Z}_N) \preceq \Pi_5$ by Theorem 3.18. This is impossible, because a subsequence of Π_5 with five terms is Π_5 itself.

The case $D_N(a) = 0$. We use the lists $\mathcal{Z}_t$ of the proof of Theorem 3.18 and number the entries of $\mathcal{Z}_t$ from 1 to $2N - t - 1$. So $D_t(b)$ stands at place $N - t + b$ for $1 \le b \le t-1$. For $3 \le t \le N-1$ let W_t be the list w of Lemma 3.16 for $\zeta = \mathcal{Z}_t$ and $\eta = \mathcal{Z}_{t+1}$. Then $\mathrm{sp}(W_t) = \mathrm{sp}(\mathcal{Z}_t) \preceq \Pi_5$. The entry of $\mathcal{Z}_{t+1}$ at place k is a combination of the entries of $\mathcal{Z}_t$ at the places k and $k+1$, with nonnegative coefficients. We call these two entries its parents. For $2 \le b \le t-1$ the parents of $D_{t+1}(b)$ are $D_t(b-1)$ and $D_t(b)$, and both coefficients are positive.

For $4 \le t \le N$ let B_t be the set of integers b with $\max(2, a - N + t) \le b \le \min(a, t-2)$. It is not empty, because $2 \le a \le N-2$. We prove by downward induction on t that there are places $p_1 < p_2 < p_4 < p_5$ in $\mathcal{Z}_t$ such that the entries there have the signs $-,+,+,-$, that $D_t(b) = 0$ for every $b \in B_t$, and that all the places $N - t + b$ with $b \in B_t$ lie strictly between p_2 and p_4. For $t = N$ we have $B_N = \{a\}$, and we take the places a_0, a_1, $N - a_1$ and $N - a_0$.

Let the claim hold for $t+1$, where $4 \le t \le N-1$, with the places $p_1' < p_2' < p_4' < p_5'$. Let $b \in B_{t+1}$. Then $D_{t+1}(b) = 0$ is a combination of its two parents with positive coefficients. So the parents both vanish, or they have strictly opposite signs. In the second case one parent is negative.

In W_t it stands next to $D_{t+1}(b)$, hence between the positive entries of $\mathcal{Z}_{t+1}$ at the places p'_2 and p'_4, and these stand between the negative entries at p'_1 and p'_5. So $(-,+,-,+,-) \preceq \mathrm{sp}(W_t) \preceq \Pi_5$, which is impossible. Hence $D_t(b-1) = D_t(b) = 0$ for every $b \in B_{t+1}$. Every element of B_t is of the form b or $b-1$ with $b \in B_{t+1}$, so D_t vanishes on B_t.

The entry of $\mathcal{Z}_{t+1}$ at p'_1 is negative, so one of its parents is negative. Let $p_1 \in \{p'_1, p'_1+1\}$ be its place in $\mathcal{Z}_t$, and choose p_2, p_4 and p_5 in the same way. Then $p_1 \le p'_1 + 1 \le p'_2 \le p_2$, and $p_1 \neq p_2$ because the signs differ. In the same way $p_4 < p_5$. Let q^- and q^+ be the smallest and the largest place $N-t-1+b$ with $b \in B_{t+1}$. The places $N-t+b$ with $b \in B_t$ lie between q^- and $q^+ + 1$. Since $p_2 \le p'_2 + 1 \le q^-$ and $p_4 \ge p'_4 \ge q^+ + 1$, and the entries at p_2 and p_4 are not zero, these places lie strictly between p_2 and p_4. This proves the claim for t.

The induction step with $t = 4$ shows that both parents of $D_5(b)$ vanish for every $b \in B_5 \subseteq \{2,3\}$. The parents of $D_5(2)$ are s_4 and $D_4(2)$, and those of $D_5(3)$ are $D_4(2)$ and s_4. So $s_4 = D_4(2) = 0$. By (2) with $t = 3$ and $a = 2$, and since $D_3(1) = D_3(2) = s_3$, we have $D_4(2) = r_3(0)s_3 + \big(1 - r_3(3)\big)s_3 = 2\varepsilon s_3$. So $s_3 = 0$ as well, which contradicts Lemma 3.17. □

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